PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Software Requirements Specification

Collaborative Study Material & Quiz Platform

Version 1.2 (draft – pending team approval)

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Revision History

Name	Date	Reason For Changes	Version
Project Team	03-10-2026	Initial release of the SRS for Part 1 deliverables	1.0
Project Team	07-10-2026	Final template-compliance and quality audit: Product Scope added, RTM completed, consistency fixes	1.0
Project Team	14-10-2026	Team expanded from three to four members: Member 4 added; Appendix D (requirement ownership matrix) added; DC-11 amended to "four-member". No requirement ID or requirement text changed.	1.1 (draft)
Project Team	17-10-2026	Final audit against the Project Guidelines, Deliverables Part-1/2, SAD and STP: team structure aligned to four members (Appendix D revised, ownership of security objectives and cross-module NFRs aligned to the STP and the confirmed roles); Appendix D.6 corrected to show the amended DC-11; SAD, STP and Part-2 added to References; TOC wording fixed; TC-QS-04 added to the SEC-ACC-04 row of Appendix C. No requirement ID or requirement text changed. Draft pending team approval.	1.2 (draft)
Project Team	18-10-2026	Re-issue of Revision 1.2 (no change of revision number): Member 4 identified as Vedh on the cover page and in Appendix D.1 (SRN still to be supplied); group number 17 entered on the cover; cover date updated; Appendix D checked against the ownership workbook (115 IDs, no difference). No requirement ID, requirement text, acceptance criterion or	1.2 (draft)

		ownership allocation changed. Draft pending team approval.	

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional, non-functional and security requirements of the Collaborative Study Material & Quiz Platform, Version 1.0. The platform lets users manage accounts, share study materials, maintain a question bank, take quizzes, and view performance and collaboration information.

This document specifies the four platform features – Account Management, Material Sharing, Question Bank & Quizzes and Performance & Collaboration (Section 3.2.3) – together with the security requirements for protected functionality. It defines user-system interactions, system responses to valid and invalid input, access control rules, security objectives and requirements traceability.

It serves as the reference for design, implementation, verification and testing.

1.2 Intended Audience

- **Students and Faculty:** end users who manage accounts and upload, share, download and manage study materials.
- **Administrators:** users responsible for platform-level management.
- **Developers:** implement the system according to the specified requirements.
- **Testers:** verify that the system satisfies the requirements and handles invalid conditions correctly.
- **Project managers:** coordinate development and ensure the deliverables are completed.
- **Reviewers:** assess the requirements, design, implementation and testing documentation.

1.3 Product Scope

The Collaborative Study Material & Quiz Platform is a web application that gives Students, Faculty and Admin users one place to share study materials, assess learning and collaborate. Its objectives are to let users share study materials under owner-controlled access, to let Faculty build a question bank and timed quizzes that are graded automatically, to give Students and Faculty performance feedback derived from quiz attempts, and to support collaborative learning through study groups with Admin moderation. Security and access control apply to every feature.

In scope: Account Management (registration, authentication, profile, logout), Material Sharing (upload, view, download, share, update, remove, revoke access), Question Bank & Quizzes (question management, quiz creation, attempts, automatic grading, results), Performance & Collaboration (personal, topic-wise and class performance, study groups, content moderation), and the security, performance, safety and quality requirements in Section 3.3.

Out of scope: question types other than single-correct multiple-choice questions; email notifications for quizzes; analytics beyond REQ-PA-01 to REQ-PA-06; creation of Admin accounts through public registration (BR-08); and anything outside the four features above (DC-09).

1.4 Definitions, Acronyms and Abbreviations

Term	Definition
SRS	Software Requirements Specification

RTM	Requirements Traceability Matrix
UML	Unified Modeling Language
UC	Use case identifier
REQ	Functional requirement identifier
NFR	Non-functional requirement identifier
SEC	Security requirement identifier
SO	Security objective identifier
Platform User	Abstract actor that generalizes Student, Faculty and Admin; any registered user of the platform
Student	A user who accesses study materials and takes quizzes
Faculty	A user who shares study materials and manages quiz content
Admin	A user responsible for platform administration and content moderation
Material Owner	The user who uploaded a study material and is recorded as its owner
Authorized User	A registered user to whom the owner has granted access to a material
Study Material	A file shared through the platform for study purposes
Protected Functionality	Functionality that requires a valid authenticated session
Session	The authenticated period during which a user can access functionality permitted for their role
TLS	Transport Layer Security, which protects data in transit between client and server
bcrypt	A password-hashing algorithm that stores passwords as salted hashes
PDF, DOCX, PPTX, PNG	Supported study material file formats
Question Bank	Shared collection of multiple-choice questions managed by Faculty

Quiz	A timed set of 1 to 50 questions created by a Faculty user from the question bank
Quiz Attempt	A Student's single timed attempt at a published quiz, graded automatically
Draft / Published	Quiz states: a Draft is visible only to its owner; a Published quiz is available to Students

Additional terms used in this document are defined in Appendix A: Glossary.

1.5 References

1. Software Engineering Mini-Project Guidelines: project requirements, team responsibilities, functional features and qualitative properties.
2. Software Engineering Mini-Project Deliverables Part-1: required SRS, SAD, Test Plan, security, traceability, diagram, API and testing deliverables.
3. PES University Software Requirements Specification Template: structure of this SRS.
4. Software Architecture and Design Specification Template: structure of the SAD.
5. Software Test Plan Template: structure of the Test Plan.
6. Software Architecture and Design Specification (SAD), Collaborative Study Material & Quiz Platform, Team 17, Version 1.0, 03-10-2026: architecture and design that realise these requirements.
7. Software Test Plan (STP), Collaborative Study Material & Quiz Platform, Team 17, Version 1.0, 03-10-2026: test cases that verify these requirements.
8. Software Engineering Mini-Project Deliverables Part-2: backlog creation from the functional and non-functional requirements of this SRS, and the Sprint 1 and Sprint 2 plan.

1.6 Overview

- **Section 1** introduces the purpose, intended audience, product scope, terminology, references and organization of this document.
- **Section 2** describes the product: perspective, functions, user classes, operating environment, constraints, and assumptions and dependencies.
- Section 3 specifies the external interface requirements (3.1), the functional requirements (3.2: analysis models, use case specifications and system features), the non-functional requirements (3.3) and the design constraints (3.4).
- The system features are Account Management (registration, authentication, profile management, logout), Material Sharing (upload, view, download, share, update, remove, revoke access), Question Bank & Quizzes (question management, quiz creation, attempts, automatic grading, results) and Performance & Collaboration.
- The security objectives, requirements and scenarios (account confidentiality, material access control, integrity, secure sessions, secure communication and assessment integrity) are specified in Section 3.3.3.
- Appendix A is the glossary, Appendix B the field layouts and Appendix C the Requirement Traceability Matrix, which links requirements to use cases, architecture, design, code and tests. Appendix D records the four-member requirement ownership matrix.

2. Overall Description

2.1 Product Perspective

The Collaborative Study Material & Quiz Platform is a new, self-contained web application for academic study-material sharing and collaborative learning. It combines account management, material sharing, a question bank with quizzes, and performance and collaboration features.

Authenticated users access functionality according to their role. Account Management, Material Sharing, Question Bank & Quizzes and Performance & Collaboration are the user-facing features specified in this SRS. The platform stores user accounts, study-material records, ownership information and access permissions, and applies security controls so that only authenticated and authorized users can perform permitted operations.

2.2 Product Functions

- **Account Management:** register, log in, view and update profile, log out.
- **Material Sharing:** upload, view, download, share, update, remove study materials, and revoke shared access.
- **Question Bank & Quizzes:** manage questions, create quizzes, attempt quizzes, automatic grading, view results.
- **Performance & Collaboration:** view personal performance, class performance and topic-wise analysis, create or join study groups, moderate content.
- **Security and Access Control:** authentication before protected functionality, role-based and ownership-based permissions, password and session protection, TLS-secured communication.

2.3 User Classes and Characteristics

User class	Characteristics and responsibilities	Importance
Student	Frequent users. Access, upload and share study materials, attempt quizzes, view performance. Basic web skills. Standard privileges.	Highest
Faculty	Regular users. Share materials, manage the question bank and quizzes, view class performance. Basic to moderate web skills. Standard privileges, with quiz-management permissions.	High
Admin	Occasional users. Platform management and content moderation. Elevated privileges.	Medium

Platform User is the common abstraction of Student, Faculty and Admin used in the use case model. **Material Owner** is a role held by the Student or Faculty user who uploaded a material; the owner can share, update, remove and revoke access to it.

All users are expected to have the basic computer and web-application skills needed to use forms, upload files and take quizzes.

2.4 Constraints

2.4.1 Operating Environment

The platform is a web application with a browser-based client and a server-side application. It requires:

- A current web browser (Chrome, Firefox, Edge or Safari) on a desktop or mobile device.
- A server to host the application and process requests.
- Persistent storage for user accounts, study-material records, access permissions and application data.
- Network connectivity with TLS 1.2 or higher for all client-server communication.
- Support for PDF, DOCX, PPTX and PNG study material files.

The platform is built on the MERN stack (DC-01); the operating system, database design, framework and deployment details are finalized in the Software Architecture and Design Specification.

2.4.2 Design and Implementation Constraints

Design and implementation constraints (DC-01 to DC-11) are specified in Section 3.4.

2.5 Assumptions and Dependencies

- Users have a supported browser and a network connection.
- Users provide accurate information when registering and managing their accounts.
- A material can be shared only with users who already have registered accounts.
- Uploaded files are stored on the user's device and meet the supported type and size limits.
- The system depends on persistent storage, the application server and secure network communication being available and working correctly.
- The detailed technology stack, database design and deployment configuration will be specified in the Software Architecture and Design Specification.

- Performance & Collaboration depends on the completed quiz attempts produced by Question Bank & Quizzes.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The system shall provide a web-based graphical user interface that exposes only the functionality permitted for the user's role.

ID	Interface	Required content and behaviour
UI-01	Registration	Fields for full name, email, password and role (Student or Faculty). Validation errors name the field and rule to correct.
UI-02	Login	Fields for email and password. Shows "Invalid email or password" on failure and a lockout notice (15 minutes) when the account is locked. Prompts re-login when a session has expired.
UI-03	Profile	Shows profile information. Allows editing of name, institution and profile picture only; email and role are read-only.
UI-04	Material Upload	Fields for title, subject, description and file selection. Shows specific errors for missing information, unsupported type or oversized file.
UI-05	Material Library	Lists only the materials the user owns or is authorized to access, with view and download options. Shows "Access denied" for unauthorized requests.
UI-06	Material Management	For owners only: update, remove (with confirmation), share and revoke access.
UI-07	Logout	A logout action available on every authenticated screen.
UI-08	Question Bank	For Faculty only. Lists questions with text, topic, difficulty and owner; search box and topic/difficulty filters; Add control; Edit and Remove controls shown only for questions the user owns.
UI-09	Question Form	Fields for question text, four options, correct option, topic, difficulty and optional explanation. Validation errors name the field and rule to correct.
UI-10	Quiz Builder	For Faculty only. Fields for title, description, time limit and availability start/end; a question picker for selecting 1 to 50 questions; Save as Draft, Publish, Edit and Delete (with confirmation) controls.
UI-11	Quiz List	For Students. Lists open and upcoming published quizzes with title, number of questions, time limit, availability window and attempted status; Start Quiz is enabled only when the quiz is open and not yet attempted.
UI-12	Quiz Attempt	Shows the questions with single-choice options, a countdown of the remaining time, an answered/unanswered indicator per question and a Submit control that warns about unanswered questions. Never shows correct answers.

UI-13	Quiz Result	Shows score, percentage and, per question, the Student's answer, the correct answer, correctness and explanation; also lists the Student's past attempts.
UI-14	Quiz Submissions	For the quiz owner only. Lists each Student's name, score, percentage and submission time for the owner's quiz.
UI-15	Performance	For Students. Shows completed quiz attempts with quiz title, score, percentage and submission date, the Student's performance information, and performance grouped by topic. Shows "No performance data available" or "No topic-wise performance data available" when no completed attempts exist.
UI-16	Class Performance	For Faculty only. After the Faculty user selects a quiz they own, lists each Student's name, score, percentage and submission information. Shows "No class performance data available" when there are no submissions and "Access denied" for a quiz the user does not own.
UI-17	Study Groups	For Students. Controls to create a group (group name and optional description), join an existing group and list the groups the Student belongs to. Informs the Student when membership already exists and shows "Group unavailable" when a group cannot be retrieved.
UI-18	Content Moderation	For Admin only. Lists collaborative content available for moderation with the available moderation actions and a confirmation step. Shows "Access denied" to non-Admin users and "Content unavailable" when content cannot be retrieved.

Interface standards:

- Error messages appear next to the relevant field or at the top of the screen, and identify the rule that failed.
- Navigation, logout and help controls are placed consistently on every authenticated screen.
- The interface adapts to desktop and mobile screen sizes.
- Interface wording for the Question Bank, Quizzes and Performance features is specified in their own feature sections.

3.1.2 Software Interfaces

The platform comprises the following software components. Detailed technologies, APIs and component interfaces are specified in the Software Architecture and Design Specification.

- **Client application:** sends user requests to the server and displays the responses.
- **Application server:** validates input, authenticates users, enforces authorization and applies business rules.
- **Authentication and session management:** issues and validates session tokens for protected requests.
- **Database:** stores user accounts, profile data, material metadata, ownership and access permissions, questions, quizzes with their question copies, attempts, selected answers and results.

- **File storage:** stores uploaded study materials and profile pictures within the file type and size limits.

- **Quiz engine:** manages the quiz life cycle (Draft, Published), the availability window, attempt start time and deadline, and enforces the time limit on the server.

- **Grading service:** compares the answers submitted for an attempt with the correct options stored in the quiz copy and computes the score and percentage (REQ-QZ-11, REQ-QZ-12).

- **Performance analytics service:** retrieves a Student's completed quiz attempts and calculates personal, topic-wise and class performance (REQ-PA-01 to REQ-PA-06).

- **Study group and collaboration service:** manages study groups, memberships and moderation of collaborative content (REQ-COL-01 to REQ-COL-07).

3.1.3 Communications Interfaces

- **COM-01:** All client-server communication shall use HTTPS with TLS 1.2 or higher, and unencrypted connections shall be rejected (SEC-COM-01).
- **COM-02:** Credentials shall be transmitted only over the encrypted connection.
- **COM-03:** Every request for protected functionality shall carry a valid session token.
- **COM-04:** The server shall return distinct responses for success, validation failure, authentication failure, authorization failure and unavailable material.
- **COM-05:** Material downloads shall be delivered only to authorized users over the encrypted connection.
- **COM-06:** Response times shall meet REQ-ACC-04, REQ-MAT-03, REQ-MAT-06, REQ-QB-03, REQ-QB-04, REQ-QZ-08 and REQ-QZ-12.

- **COM-07:** Quiz data sent to a Student shall not contain correct options or explanations until the attempt has been submitted (SEC-QZ-02).

- **COM-08:** The server clock, not the client clock, shall determine the attempt start time, deadline and acceptance of answers; the client receives the remaining time only for display (SEC-QZ-04).

- **COM-09:** The server shall return distinct responses for quiz not yet open, quiz closed, quiz already attempted, time limit expired and quiz unavailable.

- **COM-10:** The server shall return distinct responses for access denied (REQ-PA-06, REQ-COL-06), a student already being a group member (REQ-COL-03) and an unavailable study group or collaborative content item (REQ-COL-07).

3.1.4 Hardware Interfaces

The platform has no dedicated hardware interfaces. It runs on standard user devices (desktop, laptop or mobile) with a web browser.

3.2 Functional Requirements

3.2.1 Analysis Models

Figure 1 is the summary-level UML use case diagram of the whole platform. It covers Account Management, Material Sharing, Question Bank & Quizzes and Performance & Collaboration for the actors Platform User, Student, Faculty and Admin. Figure 2 shows the Question Bank & Quizzes use cases in detail with their use case IDs. The use case specifications in Section 3.2.2 are authoritative for primary actors, flows and related requirements; for example, Create / Join Study Group (UC-COL-01) is performed by a Student.

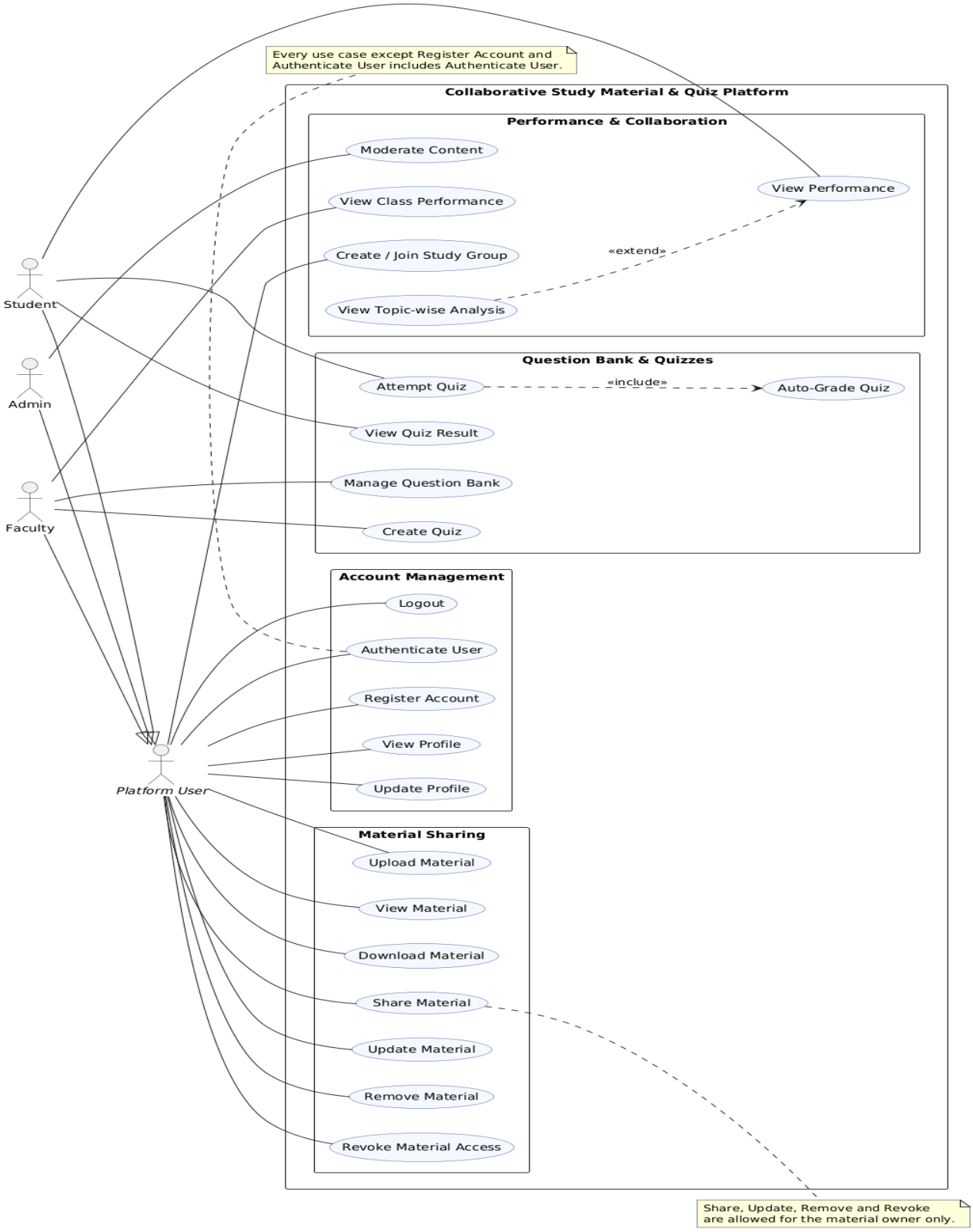


Figure 1: Use case diagram of the Collaborative Study Material & Quiz Platform (Account Management, Material Sharing, Question Bank & Quizzes, Performance & Collaboration).

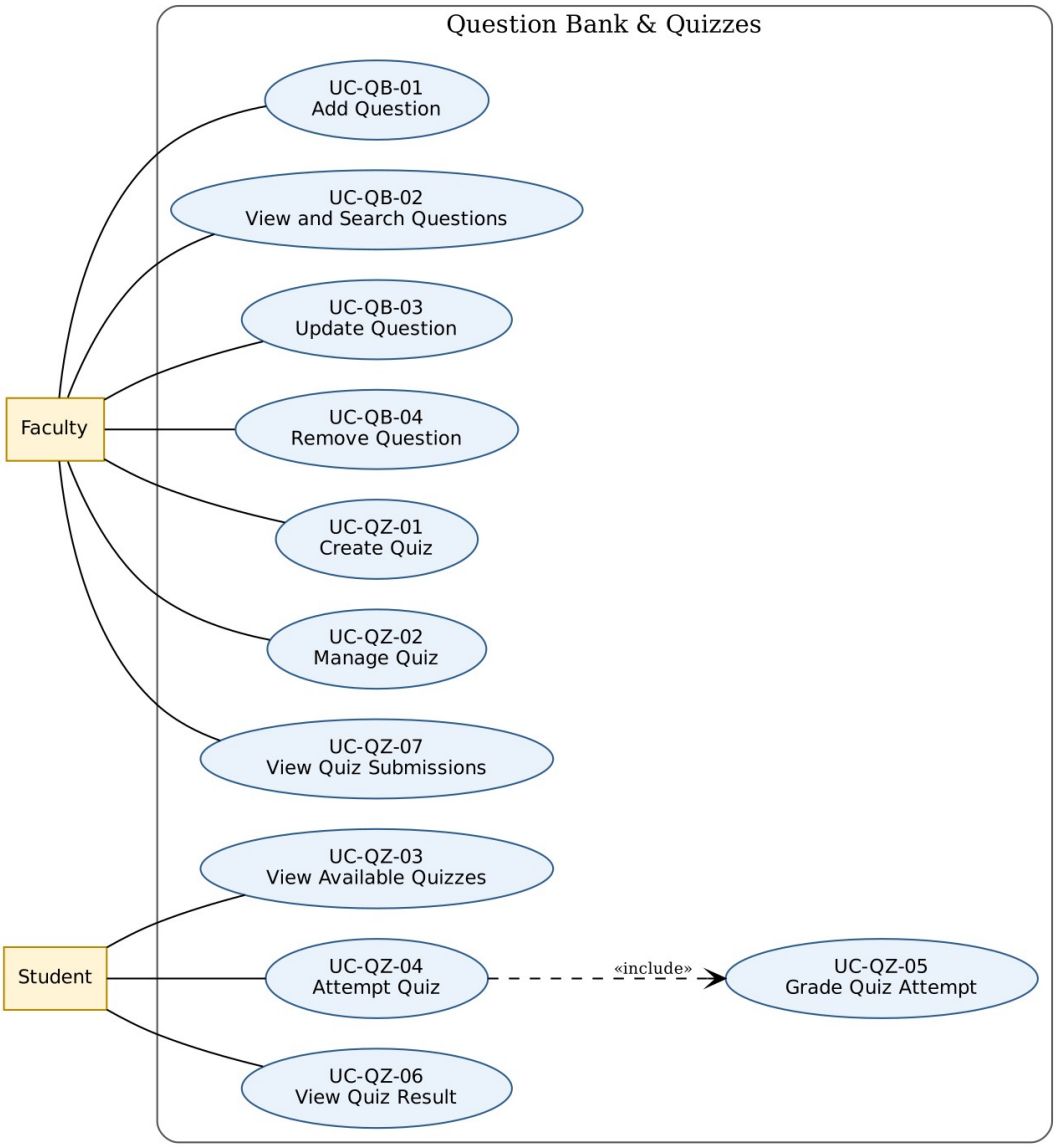


Figure 2: Detailed use cases of Question Bank & Quizzes (UC-QB-01 to UC-QB-04, UC-QZ-01 to UC-QZ-07; Faculty and Student actors).

3.2.2 Use Case Specification

3.2.2.1 Account Management

UC-ACC-01 — Register Account

Use Case ID	UC-ACC-01
Use Case Name	Register Account

Primary Actor	Prospective user (Student or Faculty)
Preconditions	The user is not authenticated and can access the registration interface.
Postconditions	A new account is created with the submitted full name, email, password and role.
Main Flow	1. User selects Register Account. 2. System displays the registration form. 3. User enters full name, email, password and role. 4. User submits the form. 5. System validates the information. 6. System creates the account. 7. System confirms successful account creation.
Alternative / Error Flows	A1. Missing information: System rejects registration and names the empty field(s). A2. Invalid information (email format or password rule): System rejects registration and names the field and rule to correct. A3. Email already registered (case-insensitive): System rejects registration and states that the email is already in use.
Related Requirements	REQ-ACC-01, REQ-ACC-02, REQ-ACC-03, SEC-ACC-03

UC-ACC-02 — Authenticate User

Use Case ID	UC-ACC-02
Use Case Name	Authenticate User
Primary Actor	Platform User (Student, Faculty, Admin)
Preconditions	The user has a registered account and is not authenticated.
Postconditions	The user is authenticated and can access functionality permitted for their role.
Main Flow	1. User selects Login. 2. System displays the login interface. 3. User enters email and password and submits. 4. System validates the credentials. 5. System authenticates the user and grants access to permitted functionality.
Alternative / Error	A1. Invalid credentials: System denies authentication and displays

Flows	"Invalid email or password". A2. Account locked (5 consecutive failed attempts): System denies authentication for 15 minutes, including for correct credentials.
Related Requirements	REQ-ACC-04, REQ-ACC-05, REQ-ACC-06; SEC-ACC-01, SEC-ACC-02, SEC-ACC-06

UC-ACC-03 — View Profile

Use Case ID	UC-ACC-03
Use Case Name	View Profile
Primary Actor	Platform User (Student, Faculty, Admin)
Preconditions	The user is authenticated.
Postconditions	The user's profile information is displayed.
Main Flow	1. User selects View Profile. 2. System verifies the authenticated session. 3. System retrieves and displays the user's profile information.
Alternative / Error Flows	A1. No valid session: System denies access and requires authentication. A2. Profile cannot be retrieved: System informs the user.
Related Requirements	REQ-ACC-08, REQ-ACC-10; SEC-ACC-01

UC-ACC-04 — Update Profile

Use Case ID	UC-ACC-04
Use Case Name	Update Profile
Primary Actor	Platform User (Student, Faculty, Admin)
Preconditions	The user is authenticated.
Postconditions	Valid changes to name, institution or profile picture are saved.
Main Flow	1. User selects Update Profile. 2. System displays the editable fields (name, institution, profile picture). 3. User modifies the fields and submits. 4. System validates the changes. 5. System saves the changes and confirms the update.

Alternative / Error Flows	A1. Empty name or picture over 2 MB: System rejects the update and names the failing field. A2. Attempt to change email, role or another user's account: System denies the change.
Related Requirements	REQ-ACC-08, REQ-ACC-09, REQ-ACC-10; SEC-ACC-05

UC-ACC-05 — Logout

Use Case ID	UC-ACC-05
Use Case Name	Logout
Primary Actor	Platform User (Student, Faculty, Admin)
Preconditions	The user has an active authenticated session.
Postconditions	The session is terminated and its token can no longer be used.
Main Flow	1. User selects Logout. 2. System terminates the session and invalidates its token. 3. System redirects the user to the unauthenticated interface.
Alternative / Error Flows	A1. Session already expired (30 minutes of inactivity): System treats the user as unauthenticated and requires login.
Related Requirements	REQ-ACC-07, REQ-ACC-10; SEC-ACC-04

3.2.2.2 Material Sharing

UC-MAT-01 — Upload Material

Use Case ID	UC-MAT-01
Use Case Name	Upload Material
Primary Actor	Platform User (Student, Faculty, Admin)
Preconditions	The user is authenticated.
Postconditions	A valid material is stored with the uploader as its owner.
Main Flow	1. User selects Upload Material. 2. System displays the upload interface. 3. User enters title, subject, description and selects a file, then submits. 4. System validates the information and file.

	5. System stores the material and records the uploader as owner. 6. System shows the material in the owner's library within 3 seconds.
Alternative / Error Flows	A1. Missing or over-length field (title above 100, description above 500 characters): System rejects the upload and names the failing field. A2. Unsupported file type (allowed: PDF, DOCX, PPTX, PNG): System rejects the upload and lists the supported types. A3. File larger than 25 MB: System rejects the upload and states the size limit.
Related Requirements	REQ-MAT-01, REQ-MAT-02, REQ-MAT-03

UC-MAT-02 — View Material

Use Case ID	UC-MAT-02
Use Case Name	View Material
Primary Actor	Platform User (Student, Faculty, Admin)
Preconditions	The user is authenticated.
Postconditions	A user with access can view the material.
Main Flow	1. User requests a study material. 2. System verifies the user is the owner or an authorized user. 3. System displays the material.
Alternative / Error Flows	A1. User is neither owner nor authorized (including after revocation): System denies access with "Access denied". A2. Material removed or cannot be retrieved: System informs the user without showing material details.
Related Requirements	REQ-MAT-04, REQ-MAT-10, REQ-MAT-11; SEC-MAT-01, SEC-MAT-04

UC-MAT-03 — Share Material

Use Case ID	UC-MAT-03
Use Case Name	Share Material
Primary Actor	Material Owner (Student, Faculty)
Preconditions	The user is authenticated and owns the selected material.
Postconditions	The selected registered user can access the material.

Main Flow	1. Owner selects an owned material. 2. System displays the sharing options. 3. Owner selects a registered user. 4. System verifies ownership. 5. System grants the user access and confirms the share.
Alternative / Error Flows	A1. Requester is not the owner: System denies the operation with "Access denied". A2. Recipient is not a registered user: System rejects the request and states that the user was not found.
Related Requirements	REQ-MAT-05, REQ-MAT-10; SEC-MAT-02

UC-MAT-04 — Download Material

Use Case ID	UC-MAT-04
Use Case Name	Download Material
Primary Actor	Platform User (Student, Faculty, Admin)
Preconditions	The user is authenticated and owns the material or has been granted access.
Postconditions	The user receives the material file.
Main Flow	1. User selects Download Material. 2. System verifies the user's permission. 3. System begins the download within 3 seconds (files up to 25 MB).
Alternative / Error Flows	A1. User is neither owner nor authorized: System denies the download with "Access denied". A2. Material removed or missing: System informs the user that it is unavailable.
Related Requirements	REQ-MAT-06, REQ-MAT-10, REQ-MAT-11; SEC-MAT-01, SEC-MAT-04

UC-MAT-05 — Update Material

Use Case ID	UC-MAT-05
Use Case Name	Update Material
Primary Actor	Material Owner (Student, Faculty)
Preconditions	The user is authenticated and owns the selected material.

Postconditions	Valid changes to the material are saved.
Main Flow	1. Owner selects an owned material and chooses Update Material. 2. Owner edits the title, subject, description or replaces the file, then submits. 3. System verifies ownership and validates the changes. 4. System saves the changes and confirms the update.
Alternative / Error Flows	A1. Requester is not the owner: System denies the operation with "Access denied". A2. Invalid information or replacement file (unsupported type or over 25 MB): System rejects the update and names the rule to correct.
Related Requirements	REQ-MAT-07, REQ-MAT-10; SEC-MAT-03

UC-MAT-06 — Remove Material

Use Case ID	UC-MAT-06
Use Case Name	Remove Material
Primary Actor	Material Owner (Student, Faculty)
Preconditions	The user is authenticated and owns the selected material.
Postconditions	The material is removed and all access to it is blocked.
Main Flow	1. Owner selects an owned material and chooses Remove Material. 2. System verifies ownership and requests confirmation. 3. Owner confirms. 4. System removes the material, blocks further access and confirms removal.
Alternative / Error Flows	A1. Requester is not the owner: System denies the operation with "Access denied". A2. Owner cancels the confirmation: System keeps the material unchanged. A3. Material already removed: System informs the user that it cannot be retrieved.
Related Requirements	REQ-MAT-08, REQ-MAT-10, REQ-MAT-11; SEC-MAT-03, SEC-MAT-04

UC-MAT-07 — Revoke Material Access

Use Case ID	UC-MAT-07

Use Case Name	Revoke Material Access
Primary Actor	Material Owner (Student, Faculty)
Preconditions	The user is authenticated, owns the selected material and has shared it with at least one user.
Postconditions	The selected user is denied access to the material from the next request onward.
Main Flow	1. Owner selects a shared material. 2. System lists the users with access. 3. Owner selects a user and chooses Revoke Access. 4. System verifies ownership. 5. System removes the user's permission and confirms the revocation.
Alternative / Error Flows	A1. Requester is not the owner: System denies the operation with "Access denied". A2. Selected user no longer has access: System informs the owner that no access exists.
Related Requirements	REQ-MAT-09, REQ-MAT-10; SEC-MAT-02, SEC-MAT-04

3.2.2.3 Question Bank

UC-QB-01 — Add Question

Use Case ID	UC-QB-01
Use Case Name	Add Question
Primary Actor	Faculty
Preconditions	The user is authenticated as Faculty.
Postconditions	A valid question is saved in the question bank and the user is recorded as its owner.
Main Flow	1. User selects Add Question. 2. System displays the question form. 3. User enters the question text, four options, the correct option, topic, difficulty and an optional explanation, and submits. 4. System validates the information. 5. Information is valid. 6. System saves the question, records the user as its owner and

	confirms.
Alternative / Error Flows	A1. Missing or invalid information: System rejects the question and names the failing field and rule (REQ-QB-02). A2. User is not Faculty: System denies the request with "Access denied".
Related Requirements	REQ-QB-01, REQ-QB-02, REQ-QB-03, REQ-QB-07; SEC-QZ-01

UC-QB-02 — View and Search Questions

Use Case ID	UC-QB-02
Use Case Name	View and Search Questions
Primary Actor	Faculty
Preconditions	The user is authenticated as Faculty.
Postconditions	The matching questions are displayed.
Main Flow	1. User selects Question Bank. 2. System displays the questions with text, topic, difficulty and owner. 3. User enters a keyword and/or selects a topic or difficulty filter. 4. System returns the matching questions within 2 seconds.
Alternative / Error Flows	A1. No question matches: System displays "No questions found". A2. User is a Student: System denies the request with "Access denied".
Related Requirements	REQ-QB-04, REQ-QB-07; SEC-QZ-01

UC-QB-03 — Update Question

Use Case ID	UC-QB-03
Use Case Name	Update Question
Primary Actor	Faculty (question owner)
Preconditions	The user is authenticated and owns the question.
Postconditions	The changes are saved in the question bank.
Main Flow	1. Owner selects Edit on a question they own. 2. System verifies ownership and displays the editable fields. 3. Owner changes the fields and submits.

	4. System validates the changes. 5. Changes are valid. 6. System saves the question and confirms the update.
Alternative / Error Flows	A1. Invalid information: System rejects the update and names the failing field and rule. A2. User is not the owner: System denies the request with "Access denied". A3. Question no longer exists: System informs the user that the question is unavailable.
Related Requirements	REQ-QB-02, REQ-QB-05, REQ-QB-07, REQ-QZ-18; SEC-QZ-01

UC-QB-04 — Remove Question

Use Case ID	UC-QB-04
Use Case Name	Remove Question
Primary Actor	Faculty (question owner)
Preconditions	The user is authenticated and owns the question.
Postconditions	The question is removed from the question bank; published quizzes are unchanged.
Main Flow	1. Owner selects Remove on a question they own. 2. System verifies ownership and asks for confirmation. 3. Owner confirms the removal. 4. System removes the question from the bank and confirms.
Alternative / Error Flows	A1. Owner cancels: System keeps the question unchanged. A2. User is not the owner: System denies the request with "Access denied".
Related Requirements	REQ-QB-06, REQ-QB-07, REQ-QZ-04; SEC-QZ-01

3.2.2.4 Quizzes

UC-QZ-01 — Create Quiz

Use Case ID	UC-QZ-01
Use Case Name	Create Quiz

Primary Actor	Faculty
Preconditions	The user is authenticated as Faculty and the question bank contains questions.
Postconditions	A new quiz is saved as Draft with the user as its owner.
Main Flow	1. User selects Create Quiz. 2. System displays the quiz form and the question picker. 3. User enters the title, description, time limit and availability window, selects 1 to 50 questions and submits. 4. System validates the information. 5. Information is valid. 6. System saves the quiz as Draft, records the user as its owner and confirms.
Alternative / Error Flows	A1. Missing or invalid information: System rejects the quiz and names the failing field and rule (REQ-QZ-02). A2. A selected question no longer exists: System rejects the quiz and names the unavailable question. A3. User is not Faculty: System denies the request with "Access denied".
Related Requirements	REQ-QZ-01, REQ-QZ-02, REQ-QZ-03, REQ-QZ-17; SEC-QZ-01

UC-QZ-02 — Manage Quiz

Use Case ID	UC-QZ-02
Use Case Name	Manage Quiz
Primary Actor	Faculty (quiz owner)
Preconditions	The user is authenticated and owns the quiz.
Postconditions	The quiz is updated, published or deleted.
Main Flow	1. Owner selects a quiz they own. 2. System verifies ownership and displays Edit, Publish and Delete options. 3. Owner selects Edit on a Draft quiz and submits valid changes. 4. System saves the changes and confirms. 5. Owner selects Publish. 6. System stores a copy of each selected question with the quiz, makes

	the quiz visible to Students and confirms. 7. Owner selects Delete and confirms. 8. System deletes the quiz and confirms.
Alternative / Error Flows	A1. Quiz has a started attempt: System rejects editing or deletion of the published quiz. A2. Invalid changes or availability end already passed: System rejects the request and names the failing field and rule. A3. User is not the owner: System denies the request with "Access denied".
Related Requirements	REQ-QZ-03, REQ-QZ-04, REQ-QZ-05, REQ-QZ-17, REQ-QZ-18; SEC-QZ-01

UC-QZ-03 — View Available Quizzes

Use Case ID	UC-QZ-03
Use Case Name	View Available Quizzes
Primary Actor	Student
Preconditions	The user is authenticated as Student.
Postconditions	The published quizzes that are still open or upcoming are displayed.
Main Flow	1. User selects Quizzes. 2. System displays published quizzes whose availability end has not passed, with title, number of questions, time limit, availability window and attempted status.
Alternative / Error Flows	A1. No quiz is available: System displays "No quizzes available". A2. No valid session: System denies access and requires authentication.
Related Requirements	REQ-QZ-06, REQ-QZ-17

UC-QZ-04 — Attempt Quiz

Use Case ID	UC-QZ-04
Use Case Name	Attempt Quiz
Primary Actor	Student
Preconditions	The user is authenticated as Student and the quiz is published.
Postconditions	The attempt is submitted (manually or automatically) with the selected

	answers and passed for grading.
Main Flow	1. User selects Start Quiz. 2. System verifies that the time is within the availability window and that the user has not attempted the quiz. 3. Attempt is allowed. 4. System records the start time and deadline and displays the questions and options without correct answers, with the remaining time. 5. User selects or changes answers. 6. System records the selected options. 7. User selects Submit. 8. System warns if any question is unanswered; on confirmation, records the submission and starts grading (UC-QZ-05).
Alternative / Error Flows	A1. Quiz is not yet open: System refuses the start and states that the quiz is not yet open. A2. Quiz is closed: System refuses the start and states that the quiz is closed. A3. User has already attempted the quiz: System refuses the start and states that the quiz was already attempted. A4. Time limit expires: System automatically submits the answers selected so far and rejects any later answer. A5. Quiz is unavailable or removed: System informs the user and displays no quiz details.
Related Requirements	REQ-QZ-07, REQ-QZ-08, REQ-QZ-09, REQ-QZ-10, REQ-QZ-14, REQ-QZ-17, REQ-QZ-18; SEC-QZ-02, SEC-QZ-04, SEC-QZ-05

UC-QZ-05 — Grade Quiz Attempt

Use Case ID	UC-QZ-05
Use Case Name	Grade Quiz Attempt
Primary Actor	System (on behalf of Student)
Preconditions	An attempt has been submitted.
Postconditions	The score and percentage are stored for the attempt and available to the Student.
Main Flow	1. Attempt is submitted. 2. System compares each selected option with the stored correct

	option of the quiz copy. 3. Comparison is complete. 4. System awards 1 mark per correct answer and 0 per incorrect or unanswered question, computes the score and percentage, stores the result and makes it available within 2 seconds.
Alternative / Error Flows	A1. Question left unanswered: System awards 0 marks for that question. A2. Grading cannot be completed: System keeps the submission unchanged, informs the Student that the result is pending and retries grading.
Related Requirements	REQ-QZ-11, REQ-QZ-12, REQ-QZ-14; SEC-QZ-05; NFR-PERF-04

UC-QZ-06 — View Quiz Result

Use Case ID	UC-QZ-06
Use Case Name	View Quiz Result
Primary Actor	Student
Preconditions	The user is authenticated as Student and has submitted an attempt.
Postconditions	The result of the selected attempt is displayed.
Main Flow	1. User selects My Results. 2. System displays the list of the user's attempts with quiz title, score, percentage and submission date. 3. User selects an attempt. 4. System verifies that the attempt belongs to the user and displays the score, percentage and, for each question, the user's answer, the correct answer, correctness and explanation.
Alternative / Error Flows	A1. Attempt belongs to another user: System denies the request with "Access denied". A2. Attempt cannot be retrieved: System informs the user and displays no result details.
Related Requirements	REQ-QZ-13, REQ-QZ-15, REQ-QZ-17, REQ-QZ-18; SEC-QZ-03

UC-QZ-07 — View Quiz Submissions

Use Case ID	UC-QZ-07
Use Case Name	View Quiz Submissions

Primary Actor	Faculty (quiz owner)
Preconditions	The user is authenticated and owns the quiz.
Postconditions	The scores of the students who attempted the quiz are displayed.
Main Flow	1. Owner selects Submissions on a quiz they own. 2. System verifies ownership. 3. Owner is authorized. 4. System displays each Student's name, score, percentage and submission time.
Alternative / Error Flows	A1. No attempt has been submitted: System displays "No submissions yet". A2. User is not the quiz owner: System denies the request with "Access denied".
Related Requirements	REQ-QZ-16, REQ-QZ-17; SEC-QZ-03

3.2.2.5 Performance and Collaboration

UC-PA-01 — View Personal Performance

Field	Content
Use Case ID	UC-PA-01
Use Case Name	View Personal Performance
Primary Actor	Student
Preconditions	The user is authenticated as a Student.
Postconditions	The Student's personal performance information is displayed.
Main Flow	1. Student selects View Performance. 2. System verifies the authenticated session. 3. System retrieves the Student's completed quiz attempts. 4. System calculates and displays the Student's performance information.
Alternative / Error Flows	A1. No completed attempts: System displays a message that no performance data is available.

Field	Content
	A2. No valid session: System denies access and requires authentication.
Related Requirements	REQ-PA-01, REQ-PA-02

UC-PA-02 — View Class Performance

Field	Content
Use Case ID	UC-PA-02
Use Case Name	View Class Performance
Primary Actor	Faculty
Preconditions	The user is authenticated as Faculty and owns the selected quiz.
Postconditions	The performance information for Students who attempted the selected quiz is displayed.
Main Flow	1. Faculty selects View Class Performance. 2. System verifies the Faculty session. 3. Faculty selects a quiz owned by the Faculty user. 4. System retrieves submitted attempts. 5. System displays Student names, scores, percentages and submission information.
Alternative / Error Flows	A1. No submissions: System displays a message that no class performance data is available. A2. User does not own the quiz: System denies the request with "Access denied".
Related Requirements	REQ-PA-04, REQ-PA-05, REQ-PA-06

UC-PA-03 — View Topic-wise Analysis

Field	Content
Use Case ID	UC-PA-03
Use Case Name	View Topic-wise Analysis
Primary Actor	Student
Preconditions	The user is authenticated as a Student.

Field	Content
Postconditions	Topic-wise performance information for the Student is displayed.
Main Flow	1. Student selects View Topic-wise Analysis. 2. System verifies the authenticated session. 3. System retrieves the Student's completed quiz results and question topics. 4. System groups the results by topic. 5. System displays the topic-wise performance information.
Alternative / Error Flows	A1. No completed quiz data: System displays a message that no topic-wise performance data is available. A2. No valid session: System denies access and requires authentication.
Related Requirements	REQ-PA-03

UC-COL-01 — Create or Join Study Group

Field	Content
Use Case ID	UC-COL-01
Use Case Name	Create or Join Study Group
Primary Actor	Student
Preconditions	The user is authenticated as a Student.
Postconditions	A new study group is created with the Student as a member, or the Student is added to an existing study group.
Main Flow	1. Student selects Study Groups. 2. System displays available group options. 3. Student selects Create Group or Join Group. 4. For Create Group, Student provides the required group information and submits it. 5. System validates the information and creates the group. 6. For Join Group, Student selects an existing group.

Field	Content
	7. System verifies that the group exists and adds the Student as a member.
Alternative / Error Flows	A1. Invalid group information: System rejects the request and identifies the failing information. A2. Group unavailable: System informs the Student that the group cannot be accessed. A3. Student is already a member: System informs the Student that membership already exists.
Related Requirements	REQ-COL-01, REQ-COL-02, REQ-COL-03, REQ-COL-04, REQ-COL-07

UC-COL-02 — Moderate Content

Field	Content
Use Case ID	UC-COL-02
Use Case Name	Moderate Content
Primary Actor	Admin
Preconditions	The user is authenticated as Admin and moderation content exists.
Postconditions	The selected moderation action is applied to the collaborative content.
Main Flow	1. Admin selects Moderate Content. 2. System verifies the Admin's privileges. 3. System displays content available for moderation. 4. Admin selects the content and moderation action. 5. Admin confirms the action. 6. System applies the moderation action and confirms the update.
Alternative / Error Flows	A1. User is not an Admin: System denies the request with "Access denied". A2. Content cannot be retrieved: System informs the Admin that the content is unavailable.
Related Requirements	REQ-COL-05, REQ-COL-06, REQ-COL-07

3.2.3 System Features

3.2.3.1 Account Management

3.2.3.1.1 Description and Priority

The **Account Management** feature provides users with the ability to create and manage their accounts and securely access the platform. It includes account registration, authentication, profile management, and logout.

The feature is assigned **High priority** because authentication and account management are required before users can access protected platform functionality, including material sharing and other user-specific features.

3.2.3.1.2 Stimulus/Response Sequences

Use Case 1: User Registration (UC-ACC-01)

Step	User Stimulus	System Response
1	User selects **Register**.	The system displays the registration form.
2	The user enters the required account information.	System accepts the entered information.
3	The user submits the registration form.	The system validates the provided information.
4	Information is valid and the account identifier is available.	The system creates the user account and confirms successful registration.
5	Required information is missing or invalid.	The system rejects the registration and identifies the information that must be corrected.
6	The account identifier already exists.	The system rejects the registration and informs the user that the identifier is already in use.

Use Case 2: User Authentication (UC-ACC-02)

Step	User Stimulus	System Response

1	User selects **Login**.	System displays the authentication interface.
2	User enters account credentials.	System receives the credentials for validation.
3	User submits the credentials.	System validates the credentials.
4	Credentials are valid.	System authenticates the user and provides access to protected functionality.
5	Credentials are invalid.	System denies authentication and displays an appropriate error message.

Use Case 3: Profile Management (UC-ACC-03, UC-ACC-04)

Step	**User Stimulus**	**System Response**
1	Authenticated user selects **Profile**.	System displays the user's profile information.
2	User modifies permitted profile information.	System accepts the changes.
3	User submits the updated information.	System validates the updated information.
4	Information is valid.	System saves the updated profile information and confirms the update.
5	Information is invalid or incomplete.	System rejects the update and identifies the required correction.

Use Case 4: User Logout (UC-ACC-05)

Step	**User Stimulus**	**System Response**
1	Authenticated user selects	System terminates the

	Logout.	user's authenticated session.
2	User attempts to access protected functionality after logout.	System denies access and requires the user to authenticate again.

3.2.3.1.3 Functional Requirements

ID	Requirement
REQ-ACC-01	The system shall allow a new user to register by providing full name, email address, password and role (Student or Faculty).
REQ-ACC-02	The system shall reject registration if any field is empty, the email format is invalid, or the email is already registered (case-insensitive), and shall name the failing field in the error message.
REQ-ACC-03	The system shall accept only passwords of at least 8 characters containing an uppercase letter, a lowercase letter and a digit.
REQ-ACC-04	The system shall authenticate a user whose email and password match a registered account and shall grant access within 2 seconds.
REQ-ACC-05	The system shall reject invalid credentials with the single message "Invalid email or password", without indicating which of the two was wrong.
REQ-ACC-06	The system shall lock an account for 15 minutes after 5 consecutive failed login attempts.
REQ-ACC-07	The system shall end a session after 30 minutes of inactivity or when the user logs out, and shall invalidate the session token.

REQ-ACC-08	The system shall allow an authenticated user to view and edit their name, institution and profile picture, but not their email or role.
REQ-ACC-09	The system shall reject profile updates with empty names or a picture larger than 2 MB, and shall name the failing field.
REQ-ACC-10	The system shall deny any request to a protected feature without a valid session and shall restrict each feature to the roles permitted for it.

3.2.3.2 Material Sharing

3.2.3.2.1 Description and Priority

The Material Sharing feature allows authenticated users to upload, view, share, download, update and remove study materials, and allows material owners to grant and revoke other users' access. It is rated **High** priority because material sharing is the platform's primary purpose.

3.2.3.2.2 Stimulus/Response Sequences

Use Case 1: Upload Study Material (UC-MAT-01)

Step	User Stimulus	System Response
1	User selects **Upload Material**.	System displays the material upload interface.
2	User provides the title, subject, description and file	System accepts the input.
3	User submits the material.	System validates the provided information and file.
4	Material is valid.	System stores the material and associates it with the uploading user as its owner.
5	Material or information is invalid.	System rejects the upload and names the failing field or rule (REQ-MAT-02).

Use Case 2: View Study Material (UC-MAT-02)

Step	User Stimulus	System Response
1	Authenticated user requests to view available study material.	System retrieves materials that the user is authorized to access.
2	User selects a material item.	System verifies the user's access permission.
3	User is authorized.	System displays the selected material or makes it available for viewing.
4	User is not authorized.	System denies access to the material and displays an appropriate message.

Use Case 3: Share Study Material (UC-MAT-03)

Step	User Stimulus	System Response
1	Material owner selects a material they own.	System displays the available sharing options.
2	Owner selects another user and requests to share the material.	System verifies that the requesting user owns the material.
3	Owner is authorized to share the material.	System grants the specified user access to the material.
4	Requesting user is not the material owner.	System denies the sharing operation.
5	Recipient is not a registered user.	The system rejects the request and states that the user was not found.

Use Case 4: Download Study Material (UC-MAT-04)

Step	User Stimulus	System Response
1	Authorized user selects **Download** for a material.	System verifies the user's access permission.
2	User is authorized.	System provides the material for download.

3	User is not authorized.	System denies the download request and displays an appropriate message.
4	Material has been removed or is missing.	System informs the user that the material is unavailable.

Use Case 5: Update or Remove Study Material (UC-MAT-05, UC-MAT-06)

Step	User Stimulus	System Response
1	Material owner selects a material they own.	System displays the available management options.
2	Owner selects **Update** or **Remove**.	System verifies that the user owns the material.
3	Owner selects **Update** and submits valid changes.	System updates the material information and confirms the update.
4	Owner selects **Remove** and confirms the action.	System removes the material and prevents further access to it.
5	User does not own the material.	System denies the requested modification or removal.
6	Updated information is invalid.	System rejects the update and identifies the required correction.

Use Case 6: Revoke Material Access (UC-MAT-07)

Step	User Stimulus	System Response
1	Material owner selects a shared material.	System displays the users currently authorized to access the material.
2	Owner selects an authorized user and chooses **Revoke**	System verifies that the requesting user is the material

	Access.	owner.
3	Owner is authorized.	System removes the selected user's access to the material.
4	Revoked user attempts to access the material.	System denies access to the material.
5	Requesting user is not the material owner.	System denies the access-revocation request.

3.2.3.2.3 Functional Requirements

Requirement ID	Requirement
REQ-MAT-01	The system shall allow an authenticated user to upload a study material by providing a title (max 100 characters), subject, description (max 500 characters) and one file.
REQ-MAT-02	The system shall accept only PDF, DOCX, PPTX and PNG files up to 25 MB, and shall reject any other upload with an error message that states the failing rule (missing field, unsupported type or file too large).
REQ-MAT-03	The system shall record the uploading user as the owner of each material and shall make the material visible in the owner's library within 3 seconds of a successful upload.
REQ-MAT-04	The system shall allow an authenticated user to view only the materials that the user owns or has been granted access to.
REQ-MAT-05	The system shall allow a material owner to share a material with another registered user by selecting that user, and shall make the material available to the recipient immediately.
REQ-MAT-06	The system shall allow a user with access to a material to download it, and shall begin the download within 3 seconds for files up to 25 MB.
REQ-MAT-07	The system shall allow a material owner to edit the title, subject and description of a material,

	and to replace its file, subject to the validation rules in REQ-MAT-02.
REQ-MAT-08	The system shall allow a material owner to remove a material after a confirmation prompt, and shall block all access to it from that point.
REQ-MAT-09	The system shall allow a material owner to revoke any user's access to a material, and shall deny that user access from the next request onward.
REQ-MAT-10	The system shall deny a request to view or download a material from any user who is neither its owner nor an authorized user, and shall deny share, update, remove and revoke requests from any user who is not its owner, with the message "Access denied".
REQ-MAT-11	The system shall inform the user when a requested material has been removed or cannot be retrieved, and shall not display any of its details.

3.2.3.3 Question Bank & Quizzes

3.2.3.3.1 Description and Priority

The Question Bank & Quizzes feature allows Faculty to build a shared bank of multiple-choice questions and to create timed quizzes from it, and allows Students to attempt published quizzes and receive automatically graded results with explanations. It includes question management, quiz creation and publication, quiz attempts with a server-enforced timer, automatic grading and result viewing.

The feature is rated High priority because it is the platform's assessment capability and the source of the data analysed by the Performance & Collaboration feature. The response-time requirement for grading (REQ-QZ-12) is the performance property owned by this feature.

3.2.3.3.2 Stimulus/Response Sequences

Use Case 1: Add Question (UC-QB-01)

Step	User Stimulus	System Response
1	User selects Add Question.	System displays the question form.
2	User enters the question text, four options, the correct option, topic, difficulty and an	System validates the information.

	optional explanation, and submits.	
3	Information is valid.	System saves the question, records the user as its owner and confirms.
4	Missing or invalid information.	System rejects the question and names the failing field and rule (REQ-QB-02).
5	User is not Faculty.	System denies the request with "Access denied".

Use Case 2: View and Search Questions (UC-QB-02)

Step	User Stimulus	System Response
1	User selects Question Bank.	System displays the questions with text, topic, difficulty and owner.
2	User enters a keyword and/or selects a topic or difficulty filter.	System returns the matching questions within 2 seconds.
3	No question matches.	System displays "No questions found".
4	User is a Student.	System denies the request with "Access denied".

Use Case 3: Update Question (UC-QB-03)

Step	User Stimulus	System Response
1	Owner selects Edit on a question they own.	System verifies ownership and displays the editable fields.
2	Owner changes the fields and submits.	System validates the changes.
3	Changes are valid.	System saves the question and confirms the update.
4	Invalid information.	System rejects the update and names the failing field and rule.
5	User is not the owner.	System denies the request with "Access denied".
6	Question no longer exists.	System informs the user that the question is unavailable.

Use Case 4: Remove Question (UC-QB-04)

Step	User Stimulus	System Response
1	Owner selects Remove on a question they own.	System verifies ownership and asks for confirmation.
2	Owner confirms the removal.	System removes the question from the bank and confirms.
3	Owner cancels.	System keeps the question unchanged.
4	User is not the owner.	System denies the request with "Access denied".

Use Case 5: Create Quiz (UC-QZ-01)

Step	User Stimulus	System Response
1	User selects Create Quiz.	System displays the quiz form and the question picker.
2	User enters the title, description, time limit and availability window, selects 1 to 50 questions and submits.	System validates the information.
3	Information is valid.	System saves the quiz as Draft, records the user as its owner and confirms.
4	Missing or invalid information.	System rejects the quiz and names the failing field and rule (REQ-QZ-02).
5	A selected question no longer exists.	System rejects the quiz and names the unavailable question.
6	User is not Faculty.	System denies the request with "Access denied".

Use Case 6: Manage Quiz (UC-QZ-02)

Step	User Stimulus	System Response
1	Owner selects a quiz they own.	System verifies ownership and displays Edit, Publish and Delete options.

2	Owner selects Edit on a Draft quiz and submits valid changes.	System saves the changes and confirms.
3	Owner selects Publish.	System stores a copy of each selected question with the quiz, makes the quiz visible to Students and confirms.
4	Owner selects Delete and confirms.	System deletes the quiz and confirms.
5	Quiz has a started attempt.	System rejects editing or deletion of the published quiz.
6	Invalid changes or availability end already passed.	System rejects the request and names the failing field and rule.
7	User is not the owner.	System denies the request with "Access denied".

Use Case 7: View Available Quizzes (UC-QZ-03)

Step	User Stimulus	System Response
1	User selects Quizzes.	System displays published quizzes whose availability end has not passed, with title, number of questions, time limit, availability window and attempted status.
2	No quiz is available.	System displays "No quizzes available".
3	No valid session.	System denies access and requires authentication.

Use Case 8: Attempt Quiz (UC-QZ-04)

Step	User Stimulus	System Response
1	User selects Start Quiz.	System verifies that the time is within the availability window and that the user has not attempted the quiz.
2	Attempt is allowed.	System records the start time and deadline and displays the questions and options without correct answers, with the remaining time.
3	User selects or changes answers.	System records the selected options.

4	User selects Submit.	System warns if any question is unanswered; on confirmation, records the submission and starts grading (UC-QZ-05).
5	Quiz is not yet open.	System refuses the start and states that the quiz is not yet open.
6	Quiz is closed.	System refuses the start and states that the quiz is closed.
7	User has already attempted the quiz.	System refuses the start and states that the quiz was already attempted.
8	Time limit expires.	System automatically submits the answers selected so far and rejects any later answer.
9	Quiz is unavailable or removed.	System informs the user and displays no quiz details.

Use Case 9: Grade Quiz Attempt (UC-QZ-05)

Step	User Stimulus	System Response
1	Attempt is submitted.	System compares each selected option with the stored correct option of the quiz copy.
2	Comparison is complete.	System awards 1 mark per correct answer and 0 per incorrect or unanswered question, computes the score and percentage, stores the result and makes it available within 2 seconds.
3	Question left unanswered.	System awards 0 marks for that question.
4	Grading cannot be completed.	System keeps the submission unchanged, informs the Student that the result is pending and retries grading.

Use Case 10: View Quiz Result (UC-QZ-06)

Step	User Stimulus	System Response
1	User selects My Results.	System displays the list of the user's attempts with quiz title, score, percentage and submission date.

2	User selects an attempt.	System verifies that the attempt belongs to the user and displays the score, percentage and, for each question, the user's answer, the correct answer, correctness and explanation.
3	Attempt belongs to another user.	System denies the request with "Access denied".
4	Attempt cannot be retrieved.	System informs the user and displays no result details.

Use Case 11: View Quiz Submissions (UC-QZ-07)

Step	User Stimulus	System Response
1	Owner selects Submissions on a quiz they own.	System verifies ownership.
2	Owner is authorized.	System displays each Student's name, score, percentage and submission time.
3	No attempt has been submitted.	System displays "No submissions yet".
4	User is not the quiz owner.	System denies the request with "Access denied".

3.2.3.3.3 Functional Requirements

Question Bank

ID	Requirement
REQ-QB-01	The system shall allow an authenticated Faculty user to add a question by providing the question text (max 500 characters), exactly four answer options (each max 200 characters), the correct option, a topic (max 50 characters), a difficulty (Easy, Medium or Hard) and an optional explanation (max 300 characters).
REQ-QB-02	The system shall reject a question if a required field is empty, a length limit is exceeded, two options are identical, or the correct option is not one of the four options, and shall name the failing field and rule in the error message.
REQ-QB-03	The system shall record the adding Faculty user as the owner of the question and shall make the question visible in the question bank within 3 seconds of saving.

REQ-QB-04	The system shall allow a Faculty user to view the question bank and to search and filter it by keyword, topic and difficulty, and shall return the results within 2 seconds for a bank of up to 1,000 questions.
REQ-QB-05	The system shall allow a question's owner to edit any of its fields, subject to the validation rules in REQ-QB-02.
REQ-QB-06	The system shall allow a question's owner to remove it after a confirmation prompt; the removal shall not change any quiz that has already been published (REQ-QZ-04).
REQ-QB-07	The system shall deny Students any access to the question bank and shall deny update and remove requests from any user who is not the question's owner, with the message "Access denied".

Quizzes

ID	Requirement
REQ-QZ-01	The system shall allow an authenticated Faculty user to create a quiz by providing a title (max 100 characters), an optional description (max 300 characters), a time limit of 5 to 180 minutes, an availability start and end date-time, and 1 to 50 questions selected from the question bank.
REQ-QZ-02	The system shall reject a quiz if the title is empty, no question is selected, more than 50 questions are selected, the time limit is outside 5 to 180 minutes, or the end date-time is not later than the start date-time, and shall name the failing field and rule.
REQ-QZ-03	The system shall save a new quiz as Draft, shall record the creating Faculty user as its owner, and shall make the quiz visible to Students only after the owner publishes it.
REQ-QZ-04	The system shall store a copy of each selected question (text, options, correct option and explanation) with the quiz when it is published, so that later edits or removal of bank questions do not affect the published quiz or any attempt.
REQ-QZ-05	The system shall allow a quiz owner to edit a Draft quiz and to delete a quiz after a confirmation prompt, and shall reject editing or deletion of a published quiz once any Student has started an attempt.
REQ-QZ-06	The system shall show a Student the published quizzes whose availability end has not passed, with title, number of questions, time limit, availability window and whether the Student has already attempted it, without showing any question or answer.
REQ-QZ-07	The system shall allow a Student to start an attempt only when the current time is

	within the quiz's availability window and the Student has not attempted that quiz before, and shall otherwise refuse with a message stating the reason (not yet open, closed, or already attempted).
REQ-QZ-08	The system shall display the quiz questions and options to the Student with the remaining time within 3 seconds of the start request, shall enforce the time limit on the server, and shall not send correct options or explanations to the client before the attempt is submitted.
REQ-QZ-09	The system shall allow a Student to select or change one option per question before submitting, shall warn the Student if any question is unanswered, and shall record the answers when the Student submits.
REQ-QZ-10	The system shall automatically submit the attempt with the answers selected so far when the time limit expires and shall reject any answer received after the time limit.
REQ-QZ-11	The system shall grade each submitted attempt automatically by awarding 1 mark for each correct answer and 0 marks for each incorrect or unanswered question, with no negative marking, and shall compute the score and the percentage to one decimal place.
REQ-QZ-12	The system shall complete grading and make the result available within 2 seconds of submission for quizzes of up to 50 questions, including when 100 attempts are submitted at the same time.
REQ-QZ-13	The system shall show the Student, after submission, the score, the percentage and, for each question, the Student's answer, the correct answer, whether the answer was correct and the explanation if one exists.
REQ-QZ-14	The system shall not allow an attempt to be changed or resubmitted after it has been submitted.
REQ-QZ-15	The system shall allow a Student to view the list of their own attempts with quiz title, score, percentage and submission date, and to open the detailed result of each.
REQ-QZ-16	The system shall allow a quiz owner to view, for the owner's own quiz, each Student's name, score, percentage and submission time.
REQ-QZ-17	The system shall allow only Faculty to create and manage quizzes and only Students to attempt them, only the quiz owner to edit, publish or delete the quiz or view its submissions, and only the attempt's owner to view its result, and shall deny every other request with "Access denied".
REQ-QZ-18	The system shall inform the user when a requested quiz, question or attempt has been removed or cannot be retrieved, and shall not display any of its details.

3.2.3.4 3.2.3.4 Performance & Collaboration

3.2.3.4.1 3.2.3.4.1 Description and Priority

The Performance & Collaboration feature allows Students and Faculty to view performance information derived from completed quiz attempts and supports collaborative learning through study groups. Students can view their personal performance and topic-wise analysis, while Faculty can view class performance for quizzes they own. The feature also allows Students to create or join study groups and allows Admin users to moderate collaborative content.

The feature is assigned High priority because performance analysis provides feedback on quiz outcomes and supports academic improvement, while study groups provide the collaboration functionality of the platform.

3.2.3.4.2 3.2.3.4.2 Stimulus/Response Sequences

Use Case 1: View Personal Performance (UC-PA-01)

Step	User Stimulus	System Response
1	Student selects **View Performance**.	System verifies that the Student has a valid authenticated session.
2	Student requests personal performance information.	System retrieves the Student's completed quiz attempts.
3	Performance data is available.	System calculates and displays the Student's performance information.
4	Student has no completed attempts.	System displays a message indicating that no performance data is available.
5	Session is invalid.	System denies access and requires authentication.

Use Case 2: View Class Performance (UC-PA-02)

Step	User Stimulus	System Response
1	Faculty selects **View Class Performance**.	System verifies that the Faculty user has a valid authenticated session.
2	Faculty selects a quiz owned by the Faculty user.	System verifies quiz ownership.
3	Faculty requests class	System retrieves the submitted attempts for the selected

Step	User Stimulus	System Response
	performance.	quiz.
4	Performance data is available.	System displays the Students' scores, percentages and submission information for the selected quiz.
5	No submissions exist.	System displays a message indicating that no class performance data is available.
6	Faculty does not own the selected quiz.	System denies the request with **"Access denied"**.

Use Case 3: View Topic-wise Analysis (UC-PA-03)

Step	User Stimulus	System Response
1	Student selects **View Topic-wise Analysis**.	System verifies that the Student has a valid authenticated session.
2	Student requests topic-wise performance.	System retrieves the Student's completed quiz results and associated question topics.
3	Topic performance data is available.	System groups the Student's results by topic and displays the corresponding performance information.
4	No completed quiz data exists.	System displays a message indicating that no topic-wise performance data is available.
5	Session is invalid.	System denies access and requires authentication.

Use Case 4: Create or Join Study Group (UC-COL-01)

Step	User Stimulus	System Response
1	Student selects **Create or Join Study Group**.	System displays the available study-group options.
2	Student chooses to create a group and provides the required group information.	System validates the provided information.
3	Group information is valid.	System creates the study group and records the Student as a member.

Step	User Stimulus	System Response
4	Student chooses to join an existing group.	System verifies that the selected study group exists and allows the Student to join.
5	Group does not exist or cannot be accessed.	System informs the Student that the study group is unavailable.
6	Student is already a member of the selected group.	System informs the Student that membership already exists.

Use Case 5: Moderate Content (UC-COL-02)

Step	User Stimulus	System Response
1	Admin selects **Moderate Content**.	System verifies the Admin's authenticated session and administrative privileges.
2	Admin selects content requiring moderation.	System retrieves the selected content and displays the available moderation action.
3	Admin confirms the moderation action.	System applies the action and records the updated content state.
4	Selected content cannot be retrieved.	System informs the Admin that the content is unavailable.
5	User is not an Admin.	System denies the request with **"Access denied"**.

3.2.3.4.3 3.2.3.4.3 Functional Requirements

ID	Requirement
REQ-PA-01	The system shall allow an authenticated Student to view their completed quiz attempts with the quiz title, score, percentage and submission date.
REQ-PA-02	The system shall calculate and display the Student's performance information from their completed quiz attempts.
REQ-PA-03	The system shall allow an authenticated Student to view their quiz performance grouped by topic.
REQ-PA-	The system shall allow an authenticated Faculty user to view performance information

ID	Requirement
04	for Students who have submitted attempts for quizzes owned by that Faculty user.
REQ-PA-05	The system shall display class performance information including Student name, score, percentage and submission information for the selected quiz.
REQ-PA-06	The system shall deny a Faculty user access to performance information for a quiz that the Faculty user does not own, with the message **"Access denied"**.
REQ-COL-01	The system shall allow an authenticated Student to create a study group by providing the required group information (group name and optional description, as listed in Appendix B.14).
REQ-COL-02	The system shall allow an authenticated Student to join an existing study group.
REQ-COL-03	The system shall prevent a Student from joining the same study group more than once.
REQ-COL-04	The system shall allow a Student to view the study groups to which the Student belongs.
REQ-COL-05	The system shall allow an authenticated Admin to moderate collaborative content.
REQ-COL-06	The system shall deny non-Admin users access to content-moderation operations with the message **"Access denied"**.
REQ-COL-07	The system shall inform the user when a requested study group or collaborative content item has been removed or cannot be retrieved, without displaying unavailable content details.

3.3 Other Nonfunctional Requirements

3.3.1 Performance Requirements

Response-time targets for login, upload visibility and download start are specified in REQ-ACC-04, REQ-MAT-03 and REQ-MAT-06 and apply here. The following additional targets apply under normal operating conditions.

ID	Requirement	Related

NFR-PERF-01	At least 95% of page loads shall complete within 3 seconds with 200 concurrent users.	All features
NFR-PERF-02	The system shall support 200 concurrent users with fewer than 1% of requests failing.	All features
NFR-PERF-03	A material share or access revocation shall take effect within 2 seconds of the owner's confirmation.	REQ-MAT-05, REQ-MAT-09
NFR-PERF-04	The system shall grade a submitted attempt and make the result available within 2 seconds for quizzes of up to 50 questions, including when 100 attempts are submitted at the same time. Rationale: students expect immediate feedback after submission.	REQ-QZ-11, REQ-QZ-12
NFR-PERF-05	The system shall return question-bank search and filter results within 2 seconds for a bank of up to 1,000 questions.	REQ-QB-04
NFR-PERF-06	The system shall display the questions of a quiz within 3 seconds of the Student starting the attempt.	REQ-QZ-08

3.3.2 Safety Requirements

The platform controls no physical devices, so no physical safety requirements apply. The following safeguards protect users from loss or corruption of data.

ID	Requirement
NFR-SAFE-01	The system shall back up user accounts, material records and files at least daily, and shall be able to restore from a backup no more than 24 hours old.
NFR-SAFE-02	If an upload, update or removal is interrupted or fails, the system shall leave the previous state of the material unchanged and shall not show the operation as successful.
NFR-SAFE-03	A downloaded file shall be identical to the file that was uploaded (verifiable by checksum).

3.3.3 Security Requirements

3.3.3.1 Security Objectives

ID	Objective
SO-01	**Account confidentiality:** Protect user credentials and account data from unauthorized disclosure.
SO-02	**Material access control:** Ensure study materials are accessible only to their owner and users the owner has authorized.
SO-03	**Integrity:** Prevent unauthorized modification of accounts, materials and material permissions.
SO-04	**Secure authentication and sessions:** Ensure protected functionality is reachable only through a valid session and that ended sessions cannot be reused.
SO-05	**Secure communication:** Protect data in transit between client and server.
SO-06	**Assessment integrity:** Ensure that quiz answers and results are disclosed only to authorized users and that attempts and scores cannot be tampered with.

3.3.3.2 Security Requirements

ID	Requirement
SEC-ACC-01	The system shall require successful authentication before granting access to any protected account or material function.
SEC-ACC-02	The system shall reject invalid credentials with the single message "Invalid email or password".
SEC-ACC-03	The system shall store passwords only as salted bcrypt hashes and shall never return them in any response or log.
SEC-ACC-04	The system shall terminate a session on logout or

	after 30 minutes of inactivity and shall reject any later request using that session token.
SEC-ACC-05	The system shall allow a user to modify only their own account information.
SEC-ACC-06	The system shall lock an account for 15 minutes after 5 consecutive failed login attempts.
SEC-MAT-01	The system shall deny access to a material when the requester is neither its owner nor an authorized user.
SEC-MAT-02	The system shall allow only a material's owner to share it or revoke access to it.
SEC-MAT-03	The system shall prevent any user other than the owner from updating or removing a material.
SEC-MAT-04	The system shall deny access to a material on the next request after access is revoked or the material is removed.
SEC-COM-01	The system shall transmit all client-server data over TLS 1.2 or higher and shall reject unencrypted connections.
SEC-QZ-01	The system shall allow only Faculty users to create, update or remove questions and quizzes.
SEC-QZ-02	The system shall not disclose correct options or explanations to a Student before that Student's attempt has been submitted.
SEC-QZ-03	The system shall allow only the owner of an attempt to view its result and only the owner of a quiz to view its submissions.
SEC-QZ-04	The system shall enforce the quiz availability window and time limit on the server and shall reject answers received after the time limit.
SEC-QZ-05	The system shall prevent modification of an attempt after submission and shall compute scores only on the server, never accepting a score supplied by the client.

3.3.3.3 Security Scenarios

Security Scenario 1: Unauthorized Account Access

Step	User/System Stimulus	System Response
1	User attempts to access protected functionality without authentication.	System checks the user's authentication status.
2	User has no valid authenticated session.	System denies access and requests authentication.
3	User provides valid credentials.	System authenticates the user and grants access according to the user's permissions.
4	User provides invalid credentials.	System denies authentication and displays an "Invalid email or password" without identifying the incorrect credential.

Security Scenario 2: Unauthorized Material Access

Step	User/System Stimulus	System Response
1	User requests access to a study material.	System verifies the user's authorization for the material.
2	User is the material owner.	System grants access.
3	User has been granted access by the owner.	System grants access.
4	User has not been granted access and is not the owner.	System denies access and displays **"Access denied"**.

Security Scenario 3: Unauthorized Material Modification

Step	User/System Stimulus	System Response
1	User requests to update, remove, share, or revoke access to a material.	System verifies whether the requesting user is the material owner.
2	User is the material owner.	System allows the requested operation.

3	User is not the material owner.	System denies the operation and displays **"Access denied"**.

Security Scenario 4: Revoked or Terminated Access

Step	User/System Stimulus	System Response
1	Material owner revokes a user's access.	System removes the user's authorization for the material.
2	The previously authorized user requests the material.	System verifies the current access permissions.
3	User's access has been revoked.	System denies access.
4	User logs out or their session expires.	System terminates the authenticated session.
5	User attempts to access protected functionality using the terminated session.	System denies access and requires authentication again.

Security Scenario 5: Assessment Integrity

Step	User/System Stimulus	System Response
1	Student requests the questions of a quiz.	System returns question text and options only, without correct options or explanations.
2	Student submits answers after the time limit has expired.	System rejects the late answers and grades only answers received before the deadline.
3	Student attempts to resubmit or modify a submitted attempt, or sends a score value.	System rejects the request; the stored attempt and the server-computed score are unchanged.
4	Another user requests the result of the attempt.	System denies access and displays "Access denied".

Security Scenario 6: Unencrypted Communication

Step	User/System Stimulus	System Response
1	A client attempts to connect without encryption, or with a TLS version older than 1.2.	System rejects the connection.

2	Client connects over HTTPS with TLS 1.2 or higher and submits credentials.	System accepts the connection; credentials are transmitted only over the encrypted connection.
3	A user downloads a material or views quiz data.	System delivers the data only over the encrypted connection.

3.3.3.4 Security Traceability

Objective	Security requirements	Functional requirements
SO-01	SEC-ACC-02, 03, 06	REQ-ACC-04, 05, 06
SO-02	SEC-MAT-01, 02, 04	REQ-MAT-04, 05, 06, 09, 10
SO-03	SEC-ACC-05, SEC-MAT-02, 03	REQ-ACC-08, REQ-MAT-07, 08, 10
SO-04	SEC-ACC-01, 04	REQ-ACC-04, 07, 10
SO-05	SEC-COM-01	All
SO-06	SEC-QZ-01, 02, 03, 04, 05	REQ-QB-07, REQ-QZ-07, 08, 10, 11, 14, 17

3.3.4 Software Quality Attributes

ID	Attribute	Requirement
NFR-QUAL-01	Usability	A first-time user shall be able to register, log in and upload a material in under 5 minutes without instructions.
NFR-QUAL-02	Availability	The platform shall be available at least 99% of the time each month, excluding announced maintenance.
NFR-QUAL-03	Robustness	Invalid input or an unsupported file shall be rejected without crashing the system or affecting other users.
NFR-QUAL-04	Portability	The platform shall work on the current versions of Chrome, Firefox, Edge and Safari on desktop and mobile devices.
NFR-QUAL-05	Maintainability (Modularity)	Account Management, Material Sharing, Question Bank & Quizzes and Performance & Collaboration shall be implemented as separate modules, each testable independently.
NFR-	Testability	Every requirement shall have a unique ID and be traceable to

QUAL-06	(Traceability)	at least one test case through the Requirement Traceability Matrix.

3.3.5 Business Rules

ID	Rule
BR-01	Protected functions are available only to authenticated users.
BR-02	Each study material has exactly one owner: the user who uploaded it.
BR-03	Only the owner may share, update, remove or revoke access to a material.
BR-04	A user may view or download a material only if they are its owner or have been granted access.
BR-05	Materials may be shared only with registered users.
BR-06	Users may change only their own profile; email address and role cannot be changed.
BR-07	Users may upload only material they have the right to share. The uploader is responsible for copyright compliance.
BR-08	Admin accounts are created by platform administrators, not through public registration.

Domain requirement: the platform handles academic study material for registered students and faculty, so all user-facing content and messages are in English.

3.3.6 Other Requirements

No database, internationalization or reuse requirements apply to the current scope. Legal responsibility for uploaded content is covered by BR-07. Requirements identified later shall be added through the project's change and traceability process.

3.4 Design Constraints

3.4.1 Technology Constraints

ID	Constraint
DC-01	The system shall be a web application built on the MERN stack (MongoDB, Express, React, Node.js).
DC-02	The client shall communicate with the server only through a defined API, specified in the Software Architecture and Design Specification.
DC-03	All client-server communication shall use HTTPS with TLS 1.2 or higher (SEC-COM-01).
DC-	Passwords shall be stored only as salted bcrypt hashes (SEC-ACC-03).

04	
DC-05	The client shall run on the browsers listed in NFR-QUAL-04.

3.4.2 File and Data Constraints

ID	Constraint
DC-06	The file types and size limits (PDF, DOCX, PPTX, PNG; 25 MB), field length limits (title 100, description 500 characters) and profile picture limit (2 MB) defined in REQ-MAT-01, REQ-MAT-02 and REQ-ACC-09 shall be enforced by the storage and validation design.

3.4.3 Security and Access-Control Constraints

ID	Constraint
DC-07	Session, role and ownership checks shall be performed on the server for every protected request, and shall not depend on client-side controls alone.
DC-08	Session expiry and account lockout (REQ-ACC-06, REQ-ACC-07) shall be enforced on the server.

3.4.4 Project Constraints

ID	Constraint
DC-09	The implementation shall stay within the scope of Account Management, Material Sharing, Question Bank & Quizzes, and Performance & Collaboration.
DC-10	The requirement IDs in this SRS shall be kept unchanged in the Software Architecture and Design Specification and the Test Plan.
DC-11	The system shall be developed by the four-member project team within the course schedule.

Appendix A: Glossary

Term	Definition
Account Lockout	Temporary denial of login to an account after 5 consecutive failed login attempts.
Account Management	The platform feature covering registration, authentication, profile management and logout.
Admin	A user responsible for platform administration and content moderation.
API	Application Programming Interface; the defined interface through which the client communicates with the server.

Attempt	A single submission of a quiz by a Student.
Authorized User	A registered user to whom the material owner has granted access to a study material.
Auto-Submit	Automatic submission of a quiz attempt with the answers selected so far when the time limit expires.
Availability Window	The start and end date-time between which a published quiz can be started.
bcrypt	A password-hashing algorithm used to store passwords as salted hashes.
Class Performance	The scores, percentages and submission information of the Students who attempted a quiz, viewed by the quiz owner.
Collaborative Content	Content generated within the collaboration features, for example within study groups, that is subject to Admin moderation.
Content Moderation	Review of collaborative content and application of a moderation action by an Admin.
DOCX	Microsoft Word document format supported for study-material uploads.
Draft / Published	Quiz states: a Draft is visible only to its owner; a Published quiz is available to Students.
Faculty	A user who shares study materials, manages the question bank and quizzes, and views class performance.
HTTPS	Hypertext Transfer Protocol Secure; HTTP carried over TLS.
Material Owner	The Student or Faculty user who uploaded a study material and is recorded as its owner.
Material Sharing	The platform feature covering uploading, viewing, downloading, sharing, updating, removing and revoking access to study materials.
MERN	The technology stack consisting of MongoDB, Express, React and Node.js.
NFR	Non-functional requirement identifier used to uniquely identify a non-functional requirement.
NFR-PERF, NFR-SAFE, NFR-QUAL	Identifier prefixes for performance, safety and quality-attribute requirements.
PDF	Portable Document Format; one of the supported study-material file formats.
Performance & Collaboration	The platform feature covering performance and topic-wise analysis, class performance, study groups and content moderation.
Platform User	An abstract actor generalizing Student, Faculty and Admin; the common role used in the use case model.
PNG	Portable Network Graphics image format supported for study-material uploads.
PPTX	Microsoft PowerPoint presentation format supported for study-material uploads.
Protected Functionality	Functionality that requires a valid authenticated session.
Question Bank	The collection of questions created and managed by Faculty from which quizzes are built.
Question Bank & Quizzes	The platform feature covering question management, quiz creation, quiz attempts, automatic grading and result viewing.
Quiz	A timed set of questions created by Faculty that Students attempt and that the system grades automatically.
Quiz Question Copy	The copy of each selected question that is stored with a quiz when it is published.
REQ	Functional requirement identifier used to uniquely identify a functional requirement.

Revoke Access	Removing a user's previously granted permission to access a study material.
RTM	Requirements Traceability Matrix; a table linking requirements to use cases, architecture, design, code and test cases.
SAD	Software Architecture and Design Specification.
SEC	Security requirement identifier used to uniquely identify a security requirement.
Session	The authenticated period during which a user can access functionality permitted for their role.
Session Token	A value issued after login that identifies an authenticated session in later requests.
SO	Security objective identifier used to uniquely identify a security objective.
SRS	Software Requirements Specification; the document defining the functional, non-functional and security requirements of the platform.
Student	A user who accesses study materials, uploads and shares materials, and takes quizzes.
Study Group	A group of registered users who collaborate within the platform.
Study Group Membership	The association between a Student and a study group, created when the Student creates or joins the group.
Study Material	A file shared through the platform for academic study purposes.
Time Limit	The allowed duration of a quiz attempt, from 5 to 180 minutes.
TLS	Transport Layer Security, which protects data transmitted between the client and server.
Topic-wise Analysis	A Student's quiz performance grouped by question topic.
UC	Use Case; a description of an interaction between a user and the system that achieves a goal. Use case IDs have the form UC-xxx-nn.
UI, COM, BR, DC	Identifier prefixes for user interface, communication interface, business rule and design constraint items.
UML	Unified Modeling Language; a standard notation used to model and represent software systems.

Appendix B: Field Layouts

B.1 Registration

Field	Length	Data Type	Description	Mandatory
Full Name	1–100	String	User's full name	Y
Email	up to 254	Email	Unique, case-insensitive, valid format	Y
Password	min 8	Password	Contains uppercase, lowercase and a digit	Y
Role	–	Selection	Student or Faculty	Y

B.2 Login

Field	Length	Data Type	Description	Mandatory
Email	up to 254	Email	Registered account email	Y
Password	–	Password	Must match the registered password	Y

B.3 Profile

Field	Length	Data Type	Description	Mandatory	Editable
Full Name	1–100	String	Must not be empty	Y	Y
Institution	up to 100	String	User's institution	N	Y
Profile Picture	max 2 MB	Image	Optional profile image	N	Y
Email	–	Email	Account email	–	N (read-only)
Role	–	Selection	Assigned role	–	N (read-only)

B.4 Material Upload / Update

Field	Length	Data Type	Description	Mandatory
Title	1–100	String	Material title	Y
Subject	up to 100	String / Selection	Subject area	Y
Description	1–500	Text	Short description	Y
File	max 25 MB	File	PDF, DOCX, PPTX or PNG	Y (on upload); optional on update

B.5 Material Sharing and Access Revocation

Field	Data Type	Description	Mandatory
Material	Selection	Must be owned by the requester	Y
Recipient / Authorized User	User selection	For sharing: a registered user. For revoking: a user who currently has access.	Y

B.6 Material Library (display fields)

Field	Data Type	Description
Title	String	Material title
Subject	String	Material subject
Owner	String	Name of the owner
Access Status	String	"Owner" or "Shared with me"

View and Download controls are shown only when the user is authorized (UI-05).

B.7 System Messages

Message	Used when	Requirement
Invalid email or password	Authentication fails	REQ-ACC-05
Account temporarily locked. Try again in 15 minutes.	Lockout is active	REQ-ACC-06
Session expired. Please log in again.	Session has timed out	REQ-ACC-07
Access denied	Unauthorized access or management attempt	REQ-MAT-10
Material unavailable	Material removed or cannot be retrieved	REQ-MAT-11
Unsupported file type. Allowed: PDF, DOCX, PPTX, PNG.	Invalid file type	REQ-MAT-02
File exceeds the 25 MB limit.	File too large	REQ-MAT-02
Validation error: [field] – [rule]	Missing or invalid input	REQ-ACC-02, REQ-ACC-09, REQ-QB-02, REQ-QZ-02
Access denied	Faculty requests performance for a quiz they do not own	REQ-PA-06
Group unavailable	Requested study group cannot be retrieved	REQ-COL-07
ALREADY_MEMBER	Student is already a member of the selected group	REQ-COL-03
Access denied	Non-Admin attempts content moderation	REQ-COL-06
Content unavailable	Requested collaborative content cannot be retrieved	REQ-COL-07
No performance data available	Student has no completed attempts	REQ-PA-01, REQ-

		PA-02
No topic-wise performance data available	Student has no topic performance data	REQ-PA-03
No class performance data available	No submissions exist for selected quiz	REQ-PA-04, REQ-PA-05
Quiz not yet open	Student starts a quiz before its availability start	REQ-QZ-07
Quiz closed	Student starts a quiz after its availability end	REQ-QZ-07
Quiz already attempted	Student starts a quiz that the Student has already attempted	REQ-QZ-07
Time limit expired	An answer is received after the quiz time limit	REQ-QZ-10
Quiz unavailable	Requested quiz, question or attempt has been removed or cannot be retrieved	REQ-QZ-18
Access denied	Student accesses the question bank, or a user without the required role or ownership manages questions, quizzes or attempts	REQ-QB-07, REQ-QZ-17

B.8 Question Bank — Question

Field	Length	Data Type	Description	Is Mandatory
Question ID	–	System generated	Unique identifier of the question	Y
Question Text	500	String	Statement of the question	Y
Option A / B / C / D	200 each	String	Four distinct answer options	Y
Correct Option	1	Enum (A–D)	The one correct option	Y
Topic	50	String	Topic tag used for search and filtering	Y
Difficulty	6	Enum	Easy, Medium or Hard	Y
Explanation	300	String	Shown to the Student after submission	N
Owner ID	–	System generated	Faculty user who added the question	Y
Created /	–	Date-time	System timestamps	Y

Updated At				

B.9 Quiz

Field	Length	Data Type	Description	Is Mandatory
Quiz ID	–	System generated	Unique identifier of the quiz	Y
Title	100	String	Name of the quiz	Y
Description	300	String	Instructions or summary	N
Time Limit	3	Numeric (minutes)	Allowed time per attempt, 5 to 180	Y
Available From	–	Date-time	Start of the availability window	Y
Available Until	–	Date-time	End of the availability window; later than Available From	Y
Status	9	Enum	Draft or Published	Y
Question Count	2	Numeric	Number of questions, 1 to 50	Y
Owner ID	–	System generated	Faculty user who created the quiz	Y

B.10 Quiz Question (copy stored at publication)

Field	Length	Data Type	Description	Is Mandatory
Quiz ID	–	System generated	Quiz the copy belongs to	Y
Position	2	Numeric	Order of the question within the quiz	Y
Question Text / Options / Correct Option / Explanation	as Question	String / Enum	Copy of the bank question at publication time	Y
Source Question ID	–	System generated	Bank question the copy was taken from	Y

B.11 Attempt and Answer

Field	Length	Data Type	Description	Is Mandatory
Attempt ID	–	System generated	Unique identifier of the attempt	Y
Quiz ID / Student ID	–	System generated	Quiz attempted and the Student who attempted it	Y
Start Time / Deadline	–	Date-time	Server time at start; start time plus time limit	Y
Submission Time	–	Date-time	Time of manual or automatic submission	N
Submission Type	6	Enum	Manual or Auto	N
Status	11	Enum	In Progress, Submitted or Graded	Y
Selected Option (per question)	1	Enum (A–D) or empty	Option chosen; empty if unanswered	N
Is Correct (per question)	1	Boolean	Set by the grading service	N
Score	2	Numeric	Number of correct answers	N
Percentage	5	Numeric (1 decimal)	Score divided by number of questions × 100	N

B.12 Quiz Report Requirements

Student Quiz Result Report	Quiz Submissions Report (Faculty)
Quiz Title	Quiz Title
Attempt Start and Submission Time	Student Name
Score and Percentage	Score
Question Text	Percentage
Student's Answer	Submission Time
Correct Answer	Submission Type (Manual / Auto)
Correct / Incorrect / Unanswered	Number of Submissions

Explanation (if present)	

B.13 Performance Report Requirements

Student Performance Report (REQ-PA-01 to REQ-PA-03)	Class Performance Report – Faculty (REQ-PA-04 to REQ-PA-06)
Quiz Title	Quiz Title
Score	Student Name
Percentage	Score
Submission Date	Percentage
Topic (topic-wise analysis only)	Submission Information

B.14 Study Group

Field	Data Type	Description	Mandatory
Group ID	System generated	Unique identifier of the study group	Y
Group Name	String	Name of the study group	Y
Description	String	Short description of the group	N
Created By	User ID	Student who created the group	Y
Created At	Date-time	Group creation timestamp	Y

B.15 Group Membership

Field	Data Type	Description	Mandatory
Group ID	System generated	Study group identifier	Y
User ID	System generated	Student member identifier	Y
Joined At	Date-time	Membership creation timestamp	Y

Appendix C: Requirement Traceability Matrix

Sl. No	Requirement ID	Brief Description of Requirement	Use Case ID	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
Account Management								
1	REQ-ACC-01	Register with full name, email, password and role (Student or Faculty).	UC-ACC-01	SAD 3.4: Auth & Session Service	SAD 4.3, 4.4, UI-01	TBD	TC-ACC-01, TC-ACC-02	TC-ACC-01, TC-ACC-02
2	REQ-ACC-02	Reject registration with empty fields, invalid email or duplicate email, naming the failing field.	UC-ACC-01	SAD 3.4: Auth & Session Service	SAD 4.3, 4.4, UI-01	TBD	TC-ACC-02, TC-ACC-03, UT-ACC-01	TC-ACC-02, TC-ACC-03
3	REQ-ACC-03	Enforce password rule: 8+ characters with uppercase, lowercase and digit.	UC-ACC-01	SAD 3.4: Auth & Session Service	SAD 4.3, 4.4, UI-01	TBD	TC-ACC-04, UT-ACC-01	TC-ACC-04
4	REQ-ACC-04	Authenticate valid credentials and grant access within 2 seconds.	UC-ACC-02	SAD 3.4: Auth & Session Service	SAD 4.3, 4.4, UI-02	TBD	TC-ACC-05, UT-ACC-02	TC-ACC-05
5	REQ-ACC-05	Reject invalid credentials with the single message "Invalid email or password".	UC-ACC-02	SAD 3.4: Auth & Session Service	SAD 4.3, 4.4, UI-02	TBD	TC-ACC-06, UT-ACC-02	TC-ACC-06
6	REQ-ACC-06	Lock account for 15 minutes after 5 consecutive failed logins.	UC-ACC-02	SAD 3.4: Auth & Session Service	SAD 4.3, 4.4, UI-02	TBD	TC-ACC-07, TC-AS-01, UT-ACC-03	TC-ACC-07, TC-AS-01
7	REQ-ACC-07	End session on logout or after 30 minutes of inactivity and invalidate its token.	UC-ACC-05	SAD 3.4: Auth & Session Service	SAD 4.3, 4.4, UI-07	TBD	TC-ACC-08, TC-AS-02, UT-ACC-04	TC-ACC-08, TC-AS-02
8	REQ-ACC-08	View and edit own name, institution and profile picture (not email or role).	UC-ACC-03, UC-ACC-04	SAD 3.4: Account Service	SAD 4.3, UI-03	TBD	TC-ACC-09, UT-MAT-04	TC-ACC-09
9	REQ-ACC-09	Reject profile updates with empty name or picture over 2 MB, naming the failing field.	UC-ACC-04	SAD 3.4: Account Service, File Storage	SAD 4.3, 4.4, UI-03	TBD	TC-ACC-10, UT-MAT-04	TC-ACC-10
10	REQ-ACC-10	Deny protected-feature requests without a valid session; restrict features by role.	UC-ACC-03 to UC-ACC-05, UC-MAT-01 to UC-MAT-07	SAD 3.9, 3.4: Auth & Session Service	SAD 4.3, 4.4	TBD	TC-ACC-11, TC-AS-03	TC-ACC-11, TC-AS-03
Material Sharing								

Sl. No	Requirement ID	Brief Description of Requirement	Use Case ID	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
11	REQ-MAT-01	Upload material with title (max 100 characters), subject, description (max 500) and one file.	UC-MAT-01	SAD 3.4: Material Service, File Storage	SAD 4.3, 4.4, UI-04	TBD	TC-MAT-01, TC-MAT-02, UT-MAT-01	TC-MAT-01, TC-MAT-02
12	REQ-MAT-02	Accept only PDF, DOCX, PPTX, PNG up to 25 MB; reject others stating the failing rule.	UC-MAT-01	SAD 3.4: Material Service, File Storage	SAD 4.3, 4.4, UI-04	TBD	TC-MAT-03, TC-MAT-04, UT-MAT-01	TC-MAT-03, TC-MAT-04
13	REQ-MAT-03	Record uploader as owner; show material in owner's library within 3 seconds.	UC-MAT-01	SAD 3.4: Material Service, File Storage	SAD 4.3, UI-05	TBD	TC-MAT-05, TC-MP-01	TC-MAT-05, TC-MP-01
14	REQ-MAT-04	View only materials the user owns or has been granted access to.	UC-MAT-02	SAD 3.4: Material Service, AccessGrant	SAD 4.3, UI-05	TBD	TC-MAT-06, TC-MS-01, UT-MAT-02	TC-MAT-06, TC-MS-01
15	REQ-MAT-05	Owner shares a material with another registered user, effective immediately.	UC-MAT-03	SAD 3.4: Material Service, AccessGrant	SAD 4.3, UI-06	TBD	TC-MAT-07, UT-MAT-03	TC-MAT-07
16	REQ-MAT-06	Authorized user downloads material; download starts within 3 seconds for files up to 25 MB.	UC-MAT-04	SAD 3.4: Material Service, AccessGrant, File Storage	SAD 4.3, 4.4, UI-05	TBD	TC-MAT-08, TC-MP-02	TC-MAT-08, TC-MP-02
17	REQ-MAT-07	Owner edits title, subject, description or replaces file, subject to REQ-MAT-02 validation.	UC-MAT-05	SAD 3.4: Material Service, File Storage	SAD 4.3, 4.4, UI-06	TBD	TC-MAT-09, TC-MS-05, UT-MAT-03	TC-MAT-09, TC-MS-05
18	REQ-MAT-08	Owner removes material after confirmation; all access is then blocked.	UC-MAT-06	SAD 3.4: Material Service, AccessGrant, File Storage	SAD 4.3, 4.4, UI-06	TBD	TC-MAT-10, TC-MS-05, UT-MAT-03	TC-MAT-10, TC-MS-05
19	REQ-MAT-09	Owner revokes a user's access; denied from the next request onward.	UC-MAT-07	SAD 3.4: AccessGrant, Material Service	SAD 4.3, UI-06	TBD	TC-MAT-11, TC-MS-02, UT-MAT-03	TC-MAT-11, TC-MS-02
20	REQ-MAT-10	Deny non-authorized viewing/downloading and non-owner sharing, updating, removing, revoking with "Access denied".	UC-MAT-02 to UC-MAT-07	SAD 3.9, 3.4: Material Service, AccessGrant, Auth & Session Service	SAD 4.3, 4.4, UI-05	TBD	TC-MAT-12, TC-MS-04, TC-MS-05	TC-MAT-12, TC-MS-04, TC-MS-05
21	REQ-MAT-11	Inform user when material is removed or unretrievable, showing	UC-MAT-02, UC-	SAD 3.4: Material Service, File Storage	SAD 4.3, 4.4, UI-05	TBD	TC-MAT-13, TC-MS-03	TC-MAT-13, TC-MS-03

Sl. No	Requirement ID	Brief Description of Requirement	Use Case ID	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
		none of its details.	**MAT-04, UC-MAT-06**					

Sl. No	Requirement ID	Brief Description of Requirement	Use Case ID	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
Security Requirements								
22	SEC-ACC-01	Require successful authentication before any protected account or material function.	All except UC-ACC-01, UC-ACC-02	SAD 3.9, 3.4: Auth & Session Service	SAD 4.3, 4.4	TBD	TC-AS-03	TC-AS-03
23	SEC-ACC-02	Reject invalid credentials without disclosing which one was wrong.	UC-ACC-02	SAD 3.9, 3.4: Auth & Session Service	SAD 4.4, UI-02	TBD	TC-ACC-06	TC-ACC-06
24	SEC-ACC-03	Store passwords only as salted bcrypt hashes; never return or log them.	UC-ACC-01	SAD 3.9, 3.4: Auth & Session Service	SAD 4.3	TBD	TC-AS-04	TC-AS-04
25	SEC-ACC-04	Terminate session on logout or 30-minute inactivity; reject later use of the token.	UC-ACC-05	SAD 3.9, 3.4: Auth & Session Service	SAD 4.3, 4.4	TBD	TC-AS-02, TC-QS-04	TC-AS-02, TC-QS-04
26	SEC-ACC-05	Allow users to modify only their own account information.	UC-ACC-04	SAD 3.9, 3.4: Account Service	SAD 4.3	TBD	TC-AS-05	TC-AS-05
27	SEC-ACC-06	Lock account for 15 minutes after 5 consecutive failed logins.	UC-ACC-02	SAD 3.9, 3.4: Auth & Session Service	SAD 4.3, 4.4	TBD	TC-AS-01	TC-AS-01
28	SEC-MAT-01	Deny material access to anyone who is neither owner nor authorized user.	UC-MAT-02, UC-MAT-04	SAD 3.9, 3.4: Material Service, AccessGrant	SAD 4.3, 4.4	TBD	TC-MS-01, UT-MAT-02	TC-MS-01
29	SEC-MAT-02	Only the owner may share a material or revoke access to it.	UC-MAT-03, UC-MAT-07	SAD 3.9, 3.4: Material Service, AccessGrant	SAD 4.3, 4.4	TBD	TC-MS-04, UT-MAT-03	TC-MS-04
30	SEC-MAT-03	Only the owner may update or remove a material.	UC-MAT-05, UC-MAT-06	SAD 3.9, 3.4: Material Service, AccessGrant	SAD 4.3, 4.4	TBD	TC-MS-05, UT-MAT-03	TC-MS-05
31	SEC-MAT-04	Deny access on the next request after revocation or removal.	UC-MAT-02, UC-MAT-04, UC-MAT-06, UC-MAT-07	SAD 3.9, 3.4: Material Service, AccessGrant	SAD 4.3, 4.4	TBD	TC-MS-02, TC-MS-03	TC-MS-02, TC-MS-03
32	SEC-	Transmit all client-	All	SAD 3.9, 3.4: Client	SAD 4.1,	TBD	TC-AS-06	TC-AS-06

Sl. No	Requirement ID	Brief Description of Requirement	Use Case ID	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
	COM-01	**server data over TLS 1.2 or higher; reject unencrypted connections.**		**application, Auth & Session Service**	**4.3**			

Sl. No	Requirement ID	Brief Description of Requirement	Use Case ID	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
Question Bank & Quizzes								
33	REQ-QB-01	Add question with text, four options, correct option, topic, difficulty, optional explanation.	UC-QB-01	SAD 3.4: Question Bank Service	SAD 4.3, 4.4, UI-09	TBD	TC-QB-01, TC-QB-03	TC-QB-01, TC-QB-03
34	REQ-QB-02	Reject invalid question, naming the failing field and rule.	UC-QB-01, UC-QB-03	SAD 3.4: Question Bank Service	SAD 4.3, 4.4, UI-09	TBD	TC-QB-02, TC-QB-03, TC-QB-04, TC-QB-06, TC-QS-03, UT-QZ-02	TC-QB-02, TC-QB-03, TC-QB-04, TC-QB-06, TC-QS-03
35	REQ-QB-03	Record owner; show question in bank within 3 seconds.	UC-QB-01	SAD 3.4: Question Bank Service	SAD 4.3, UI-08	TBD	TC-QB-01, TC-QP-04	TC-QB-01, TC-QP-04
36	REQ-QB-04	View, search and filter question bank; results within 2 seconds for 1,000 questions.	UC-QB-02	SAD 3.4: Question Bank Service	SAD 4.3, UI-08	TBD	TC-QB-05, TC-QP-02	TC-QB-05, TC-QP-02
37	REQ-QB-05	Owner edits question, subject to REQ-QB-02 validation.	UC-QB-03	SAD 3.4: Question Bank Service	SAD 4.3, 4.4, UI-09	TBD	TC-QB-06	TC-QB-06
38	REQ-QB-06	Owner removes question after confirmation; published quizzes unaffected.	UC-QB-04	SAD 3.4: Question Bank Service	SAD 4.3, 4.4, UI-08	TBD	TC-QB-07	TC-QB-07
39	REQ-QB-07	Deny Students bank access and non-owner update/remove with "Access denied".	UC-QB-01 to UC-QB-04	SAD 3.4: Question Bank Service, Auth & Session Service	SAD 4.3, 4.4	TBD	TC-QB-08, TC-QB-09	TC-QB-08, TC-QB-09
40	REQ-QZ-01	Create quiz with title, description, time limit, window and 1-50 questions.	UC-QZ-01	SAD 3.4: Quiz Engine, Question Bank Service	SAD 4.3, 4.4, UI-10	TBD	TC-QZ-01, TC-QZ-03	TC-QZ-01, TC-QZ-03
41	REQ-QZ-02	Reject invalid quiz, naming the failing field and rule.	UC-QZ-01	SAD 3.4: Quiz Engine, Question Bank Service	SAD 4.3, 4.4, UI-10	TBD	TC-QZ-02, TC-QZ-03, UT-QZ-03	TC-QZ-02, TC-QZ-03
42	REQ-QZ-03	Save quiz as Draft; visible to Students only after publishing.	UC-QZ-01, UC-QZ-02	SAD 3.4: Quiz Engine	SAD 4.3, UI-10	TBD	TC-QZ-01, TC-QZ-04	TC-QZ-01, TC-QZ-04
43	REQ-QZ-04	Copy selected questions into the quiz at publication.	UC-QB-04, UC-QZ-02	SAD 3.4: Quiz Engine, Question Bank Service	SAD 4.3	TBD	TC-QB-07, TC-QZ-04	TC-QB-07, TC-QZ-04
44	REQ-	Edit Draft, delete with	UC-QZ-	SAD 3.4: Quiz	SAD 4.3,	TBD	TC-QZ-05	TC-QZ-

Sl. No	Requirement ID	Brief Description of Requirement	Use Case ID	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
	QZ-05	**confirmation; reject once an attempt has started.**	**02**	**Engine**	**4.4, UI-10**			**05**

Sl. No	Requirement ID	Brief Description of Requirement	Use Case ID	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
45	REQ-QZ-06	Student lists open and upcoming published quizzes without questions or answers.	UC-QZ-03	SAD 3.4: Quiz Engine	SAD 4.3, UI-11	TBD	TC-QZ-06	TC-QZ-06
46	REQ-QZ-07	Start attempt only inside the window and only once per quiz.	UC-QZ-04	SAD 3.4: Quiz Engine	SAD 4.3, 4.4, UI-11	TBD	TC-QS-01, TC-QZ-07, UT-QZ-04	TC-QS-01, TC-QZ-07
47	REQ-QZ-08	Show questions and remaining time; server-side timer; no answers sent before submission.	UC-QZ-04	SAD 3.4: Quiz Engine	SAD 4.3, UI-12	TBD	TC-QP-03, TC-QZ-08	TC-QP-03, TC-QZ-08
48	REQ-QZ-09	Select/change one option per question; warn on unanswered; record on submit.	UC-QZ-04	SAD 3.4: Quiz Engine	SAD 4.3, UI-12	TBD	TC-QZ-09	TC-QZ-09
49	REQ-QZ-10	Auto-submit at time limit; reject late answers.	UC-QZ-04	SAD 3.4: Quiz Engine	SAD 4.3, 4.4, UI-12	TBD	TC-QZ-10, UT-QZ-05	TC-QZ-10
50	REQ-QZ-11	Auto-grade: 1 mark per correct answer, no negative marking; score and percentage.	UC-QZ-05	SAD 3.4: Grading Service, Quiz Engine	SAD 4.3	TBD	TC-QZ-11, UT-QZ-01	TC-QZ-11
51	REQ-QZ-12	Grade and show result within 2 seconds (50 questions, 100 concurrent submissions).	UC-QZ-05	SAD 3.4: Grading Service, Quiz Engine	SAD 4.3	TBD	TC-QP-01	TC-QP-01
52	REQ-QZ-13	Show score, percentage and per-question answers, correct answers, explanation.	UC-QZ-06	SAD 3.4: Grading Service, Quiz Engine	SAD 4.3, UI-13	TBD	TC-QZ-12	TC-QZ-12
53	REQ-QZ-14	Submitted attempt cannot be changed or resubmitted.	UC-QZ-04, UC-QZ-05	SAD 3.4: Quiz Engine	SAD 4.3, 4.4	TBD	TC-QZ-13, UT-QZ-05	TC-QZ-13
54	REQ-QZ-15	Student views own attempts and detailed results.	UC-QZ-06	SAD 3.4: Quiz Engine	SAD 4.3, UI-13	TBD	TC-QZ-14	TC-QZ-14
55	REQ-QZ-16	Quiz owner views each Student's score, percentage and submission time.	UC-QZ-07	SAD 3.4: Quiz Engine	SAD 4.3, UI-14	TBD	TC-QZ-15	TC-QZ-15
56	REQ-QZ-17	Role- and ownership-based access to quizzes, attempts and results.	UC-QZ-01 to UC-QZ-04, UC-QZ-	SAD 3.4: Auth & Session Service, Quiz Engine	SAD 4.3, 4.4	TBD	TC-QZ-16, TC-QZ-17	TC-QZ-16, TC-QZ-17

Sl. No	Requirement ID	Brief Description of Requirement	Use Case ID	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
			06, UC-QZ-07					

Sl. No	Requirement ID	Brief Description of Requirement	Use Case ID	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
57	**REQ-QZ-18**	**Inform user when a quiz, question or attempt is unavailable; show no details.**	**UC-QB-03, UC-QZ-02, UC-QZ-04, UC-QZ-06**	**SAD 3.4: Quiz Engine, Question Bank Service**	**SAD 4.3, 4.4**	**TBD**	**TC-QZ-18**	**TC-QZ-18**
58	**SEC-QZ-01**	**Only Faculty create, update or remove questions and quizzes.**	**UC-QB-01 to UC-QB-04, UC-QZ-01, UC-QZ-02**	**SAD 3.9, 3.4: Auth & Session Service, Question Bank Service, Quiz Engine**	**SAD 4.3, 4.4**	**TBD**	**TC-QB-08, TC-QB-09, TC-QS-03, TC-QZ-17**	**TC-QB-08, TC-QB-09, TC-QS-03, TC-QZ-17**
59	**SEC-QZ-02**	**No correct options or explanations to a Student before submission.**	**UC-QZ-04**	**SAD 3.9, 3.4: Quiz Engine**	**SAD 4.3, 4.4**	**TBD**	**TC-QZ-06, TC-QZ-08, UT-QZ-06**	**TC-QZ-06, TC-QZ-08**
60	**SEC-QZ-03**	**Only the attempt owner views a result; only the quiz owner views submissions.**	**UC-QZ-06, UC-QZ-07**	**SAD 3.9, 3.4: Quiz Engine, Auth & Session Service**	**SAD 4.3, 4.4**	**TBD**	**TC-QS-04, TC-QZ-16**	**TC-QS-04, TC-QZ-16**
61	**SEC-QZ-04**	**Server enforces availability window and time limit.**	**UC-QZ-04**	**SAD 3.9, 3.4: Quiz Engine**	**SAD 4.3, 4.4**	**TBD**	**TC-QS-01, TC-QZ-10**	**TC-QS-01, TC-QZ-10**
62	**SEC-QZ-05**	**Attempts immutable after submission; scores computed only by the server.**	**UC-QZ-04, UC-QZ-05**	**SAD 3.9, 3.4: Quiz Engine, Grading Service**	**SAD 4.3, 4.4**	**TBD**	**TC-QS-02, TC-QZ-13**	**TC-QS-02, TC-QZ-13**
Performance & Collaboration								
63	REQ-PA-01	Authenticated Student views completed attempts with quiz title, score, percentage and submission date.	UC-PA-01	**SAD 3.4: Performance Analytics Service**	**SAD 4.3, UI-15**	**TBD**	**TC-PA-01**	**TC-PA-01**
64	REQ-PA-02	Calculate and display Student performance from completed quiz attempts.	UC-PA-01	**SAD 3.4: Performance Analytics Service**	**SAD 4.3, UI-15**	**TBD**	**TC-PA-02, UT-PA-01**	**TC-PA-02**
65	REQ-PA-03	Allow an authenticated Student to view quiz performance grouped by topic.	UC-PA-03	**SAD 3.4: Performance Analytics Service**	**SAD 4.3, UI-15**	**TBD**	**TC-PA-03, UT-PA-02**	**TC-PA-03**
66	REQ-PA-04	Faculty views performance for Students who submitted attempts for quizzes owned by that Faculty.	UC-PA-02	**SAD 3.4: Performance Analytics Service**	**SAD 4.3, UI-16**	**TBD**	**TC-PA-04, TC-PA-06, UT-PA-03**	**TC-PA-04, TC-PA-06**

Sl. No	Requirement ID	Brief Description of Requirement	Use Case ID	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
67	REQ-PA-05	Display class performance including Student name, score, percentage and submission information.	UC-PA-02	**SAD 3.4: Performance Analytics Service**	**SAD 4.3, UI-16**	**TBD**	**TC-PA-05**	**TC-PA-05**
68	REQ-PA-06	Deny Faculty access to performance for a quiz not owned by that Faculty with "Access denied".	UC-PA-02	**SAD 3.4: Performance Analytics Service, Auth & Session Service**	**SAD 4.3, 4.4, UI-16**	**TBD**	**TC-PA-06, UT-PA-03**	**TC-PA-06**
69	REQ-COL-01	Student creates a study group with required group information.	UC-COL-01	**SAD 3.4: Study Group & Collaboration Service**	**SAD 4.3, 4.4, UI-17**	**TBD**	**TC-COL-01, UT-COL-01**	**TC-COL-01**
70	REQ-COL-02	Student joins an existing study group.	UC-COL-01	**SAD 3.4: Study Group & Collaboration Service**	**SAD 4.3, 4.4, UI-17**	**TBD**	**TC-COL-02**	**TC-COL-02**
71	REQ-COL-03	Prevent duplicate study-group membership.	UC-COL-01	**SAD 3.4: Study Group & Collaboration Service**	**SAD 4.3, 4.4, UI-17**	**TBD**	**TC-COL-03, UT-COL-02**	**TC-COL-03**
72	REQ-COL-04	Student views the study groups to which they belong.	UC-COL-01	**SAD 3.4: Study Group & Collaboration Service**	**SAD 4.3, UI-17**	**TBD**	**TC-COL-04**	**TC-COL-04**
73	REQ-COL-05	Admin moderates collaborative content.	UC-COL-02	**SAD 3.4: Study Group & Collaboration Service**	**SAD 4.3, 4.4, UI-18**	**TBD**	**TC-COL-05, UT-COL-03**	**TC-COL-05**
74	REQ-COL-06	Deny non-Admin moderation with "Access denied".	UC-COL-02	**SAD 3.4: Study Group & Collaboration Service, Auth & Session Service**	**SAD 4.3, 4.4, UI-18**	**TBD**	**TC-COL-06, UT-COL-03**	**TC-COL-06**
75	REQ-COL-07	Inform user when a study group or collaborative content is removed/unavailable without displaying unavailable details.	UC-COL-01, UC-COL-02	**SAD 3.4: Study Group & Collaboration Service**	**SAD 4.3, 4.4, UI-17, UI-18**	**TBD**	**TC-COL-07**	**TC-COL-07**
Non-Functional Requirements								
76	**NFR-PERF-01**	**95% of page loads within 3 seconds with 200 concurrent users.**	**All**	**SAD 3.1, 3.5: all components (layered architecture)**	**SAD 4.1, 4.4**	**TBD**	**TC-AP-01**	**TC-AP-01**

Sl. No	Requirement ID	Brief Description of Requirement	Use Case ID	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
77	NFR-PERF-02	Support 200 concurrent users with fewer than 1% failed requests.	All	SAD 3.1, 3.5: all components (layered architecture)	SAD 4.1, 4.4	TBD	TC-AP-02	TC-AP-02
78	NFR-PERF-03	Material share or revocation takes effect within 2 seconds.	UC-MAT-03, UC-MAT-07	SAD 3.4: Material Service, AccessGrant	SAD 4.3	TBD	TC-MP-03	TC-MP-03
79	NFR-PERF-04	Grading complete and result available within 2 seconds.	UC-QZ-05	SAD 3.4: Grading Service	SAD 4.3	TBD	TC-QP-01	TC-QP-01
80	NFR-PERF-05	Question-bank search within 2 seconds for up to 1,000 questions.	UC-QB-02	SAD 3.4: Question Bank Service	SAD 4.3, UI-08	TBD	TC-QP-02	TC-QP-02
81	NFR-PERF-06	Quiz questions displayed within 3 seconds of Start Quiz.	UC-QZ-04	SAD 3.4: Quiz Engine	SAD 4.3, UI-12	TBD	TC-QP-03	TC-QP-03
82	NFR-SAFE-01	Daily backups; restore from a backup no more than 24 hours old.	All	SAD 3.4: Database, File Storage	SAD 4.4	TBD	TC-AN-01	TC-AN-01
83	NFR-SAFE-02	Interrupted upload, update or removal leaves the previous material state unchanged.	UC-MAT-01, UC-MAT-05, UC-MAT-06	SAD 3.4: Material Service, File Storage	SAD 4.4	TBD	TC-AN-02	TC-AN-02
84	NFR-SAFE-03	Downloaded file is identical to the uploaded file (checksum).	UC-MAT-04	SAD 3.4: File Storage, Material Service	SAD 4.3	TBD	TC-AN-03	TC-AN-03
85	NFR-QUAL-01	First-time user can register, log in and upload within 5 minutes without instructions.	UC-ACC-01, UC-ACC-02, UC-MAT-01	SAD 3.4: Client application	SAD 4.5	TBD	TC-AN-04	TC-AN-04
86	NFR-QUAL-02	Availability of at least 99% per month, excluding announced maintenance.	All	SAD 3.1, 3.7: all components (availability goal, redundancy)	SAD 4.4	TBD	TC-AN-07	TC-AN-07
87	NFR-QUAL-03	Invalid input or unsupported files rejected without crashing or affecting other users.	UC-ACC-01, UC-MAT-01, UC-MAT-05	SAD 3.4: all services (input validation)	SAD 4.4	TBD	TC-AN-05	TC-AN-05
88	NFR-QUAL-	Works on current Chrome, Firefox, Edge	All	SAD 3.4: Client application	SAD 4.5	TBD	TC-AN-06	TC-AN-06

Sl. No	Requirement ID	Brief Description of Requirement	Use Case ID	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
	04	**and Safari on desktop and mobile.**						
89	**NFR-QUAL-05**	**Features implemented as separate, independently testable modules.**	**All**	**SAD 3.3, 3.5: component diagram and layered architecture**	**SAD 4.1**	**TBD**	**TC-AN-08**	**TC-AN-08**
90	**NFR-QUAL-06**	**Every requirement has a unique ID and is traceable to at least one test case.**	**All**	**SAD 3.8: Traceability to Requirements**	**SAD 4.1**	**TBD**	**TC-AN-09**	**TC-AN-09**

Notes: Architecture Reference gives the component in SAD 3.4 (or the SAD section for cross-cutting items); Design Reference gives SAD sections 4.1 Design Overview, 4.3 API Design, 4.4 Error Handling, Logging & Monitoring and 4.5 UX Design, and the SRS interface ID (UI-nn). Test Case ID lists the Test Plan test cases (TC-*) and unit-test candidates (UT-*) mapped to the requirement in Test Plan Section 13; System Test Case ID lists the system-level test cases (TC-*) only. Code File Reference remains TBD until implementation starts; no code file names are assumed.

Appendix D: Requirement Ownership Matrix

This appendix records the four-member team, each member's module responsibility and the ownership of every requirement in this SRS. It does not add, remove or change any requirement, requirement ID, acceptance criterion, performance threshold, validation rule, security behaviour or error message.

The allocation is provisional and subject to team approval. Ownership means responsibility for implementing and verifying a requirement (or, for security objectives, business rules and cross-module non-functional requirements, for coordination); it is not an assignment of document-editing duties, and this SRS does not assign document sections to individual contributors.

D.1 Team Composition and Module Responsibility

Member	Name – SRN	Module responsibility	Requirement groups (primary)
Member 1	Jagadeesh – PES1UG24CS347	Account Management, authentication, session management, access-control middleware and security coordination.	REQ-ACC-01 to REQ-ACC-10; SEC-ACC-01 to SEC-ACC-06, SEC-COM-01; SO-01 to SO-06; NFR-PERF-01, 02, NFR-SAFE-01, NFR-QUAL-01, 03, 04; BR-01, 06, 08
Member 2	Alifiya – PES1UG25CD841	Question Bank and Quiz Authoring.	REQ-QB-01 to REQ-QB-07; REQ-QZ-01 to 05, 16, 17, 18; SEC-QZ-01, 03; NFR-PERF-05
Member 3	Pushkar – PES1UG24CS350	Performance Analytics and Collaboration, including study groups and content moderation.	REQ-PA-01 to REQ-PA-06; REQ-COL-01 to REQ-COL-07; NFR-QUAL-02, 05, 06
Member 4	Vedh – [SRN: not yet supplied]	Material Management and Sharing, File Storage, Quiz Execution, attempts, timing, submission and grading.	REQ-MAT-01 to REQ-MAT-11; REQ-QZ-06 to 15; SEC-MAT-01 to 04, SEC-QZ-02, 04, 05; NFR-PERF-03, 04, 06, NFR-SAFE-02, 03; BR-02 to 05, 07

Member 4's SRN is a placeholder until supplied by the team; no SRN has been assumed.

Qualitative property per member (taken from existing requirements only; proposed for team confirmation): Member 1 – confidentiality of account credentials and sessions (SEC-ACC-03, SEC-ACC-04); Member 2 – question-bank search within 2 seconds for up to 1,000 questions (NFR-PERF-05); Member 3 – features implemented as separate, independently testable modules (NFR-QUAL-05); Member 4 – downloaded file identical to the uploaded file, checked by checksum (NFR-SAFE-03). No requirement is added or changed.

D.2 Ownership Conventions

• Primary owner: exactly one member per requirement; accountable for implementation and verification, or, for security objectives, business rules and cross-module non-functional requirements, for coordination.

• Supporting: members who must collaborate because the requirement depends on their module or crosses module boundaries. Cross-module requirements remain a shared responsibility of all named members; the primary owner coordinates.

• Basis: functional ownership follows the requirement's actual description and the SAD component that realises it; ownership of cross-module non-functional and security items follows the Test Plan (STP v1.0, Section 9) allocation where it exists, adjusted for the move of Material Sharing and Quiz Execution to Member 4; security objectives are coordinated by Member 1 as security coordinator.

• Related IDs: the security (SEC, SO), quality/performance/safety (NFR), business-rule (BR) and design-constraint (DC) IDs that bear on the requirement. SEC-COM-01 / SO-05 (TLS) apply to all requirements.

D.3 Requirement Ownership Matrix

ID	Existing requirement title / short description	Primary	Supporting	Related security / quality IDs
Functional requirements – Account Management				
REQ-ACC-01	Register with full name, email, password and role (Student or Faculty).	M1	—	SEC-ACC-03, BR-08, NFR-QUAL-01, NFR-QUAL-03
REQ-ACC-02	Reject registration with empty fields, invalid email or duplicate email, naming the failing field.	M1	—	NFR-QUAL-03
REQ-ACC-03	Enforce password rule: 8+ characters with uppercase, lowercase and digit.	M1	—	SEC-ACC-03
REQ-ACC-04	Authenticate valid credentials and grant access within 2 seconds.	M1	—	SEC-ACC-01, SO-01, SO-04, NFR-QUAL-01
REQ-ACC-05	Reject invalid credentials with the single message "Invalid email or password".	M1	—	SEC-ACC-02, SO-01
REQ-ACC-06	Lock account for 15 minutes after 5 consecutive failed logins.	M1	—	SEC-ACC-06, SO-01, DC-08
REQ-ACC-07	End session on logout or after 30 minutes of inactivity and invalidate its token.	M1	—	SEC-ACC-04, SO-04, DC-08
REQ-ACC-08	View and edit own name, institution and profile picture (not email or role).	M1	—	SEC-ACC-05, SO-03, BR-06
REQ-ACC-09	Reject profile updates with empty name or picture over 2 MB, naming the failing field.	M1	M4	SEC-ACC-05, DC-06, NFR-QUAL-03
REQ-ACC-10	Deny protected-feature requests without a valid session; restrict features by role.	M1	M2, M3, M4	SEC-ACC-01, SO-04, BR-01, DC-07
Functional requirements – Material Sharing				
REQ-MAT-01	Upload material with title (max 100 characters), subject, description (max 500) and one file.	M4	—	NFR-QUAL-01, NFR-SAFE-02, BR-07, DC-06
REQ-MAT-02	Accept only PDF, DOCX, PPTX, PNG up to 25 MB; reject others stating the failing rule.	M4	—	NFR-QUAL-03, DC-06
REQ-MAT-03	Record uploader as owner; show material in owner's library within 3 seconds.	M4	—	BR-02, NFR-PERF-01
REQ-MAT-04	View only materials the user owns or has been granted access to.	M4	M1	SEC-MAT-01, SO-02, BR-04
REQ-MAT-05	Owner shares a material with another registered user, effective immediately.	M4	M1	SEC-MAT-02, SO-02, BR-03, BR-05, NFR-PERF-03
REQ-MAT-06	Authorized user downloads material; download starts within 3 seconds for files up to 25 MB.	M4	—	SEC-MAT-01, SO-02, BR-04, NFR-SAFE-03
REQ-MAT-07	Owner edits title, subject, description or replaces file, subject to REQ-MAT-02 validation.	M4	—	SEC-MAT-03, SO-03, BR-03, NFR-SAFE-02
REQ-MAT-08	Owner removes material after confirmation; all access is then blocked.	M4	—	SEC-MAT-03, SEC-MAT-04, SO-03, BR-03, NFR-SAFE-02
REQ-MAT-09	Owner revokes a user's access; denied from the next request onward.	M4	—	SEC-MAT-02, SEC-MAT-04, SO-02, BR-03, NFR-PERF-03
REQ-MAT-10	Deny non-authorized viewing/downloading and non-owner sharing, updating, removing, revoking with "Access denied".	M4	M1	SEC-MAT-01, SEC-MAT-02, SEC-MAT-03, SO-02, SO-03, BR-03, BR-04
REQ-MAT-11	Inform user when material is removed or unretrievable, showing none of its details.	M4	—	SEC-MAT-04
Functional requirements – Question Bank				
REQ-QB-01	Add question with text, four options, correct option, topic, difficulty, optional explanation.	M2	—	SEC-QZ-01
REQ-QB-02	Reject invalid question, naming the failing field and rule.	M2	—	SEC-QZ-01, NFR-QUAL-03
REQ-QB-03	Record owner; show question in bank within 3 seconds.	M2	—	SEC-QZ-01
REQ-QB-04	View, search and filter question bank; results within 2 seconds for 1,000 questions.	M2	—	NFR-PERF-05

ID	Existing requirement title / short description	Primary	Supporting	Related security / quality IDs
REQ-QB-05	Owner edits question, subject to REQ-QB-02 validation.	M2	—	SEC-QZ-01
REQ-QB-06	Owner removes question after confirmation; published quizzes unaffected.	M2	—	SEC-QZ-01
REQ-QB-07	Deny Students bank access and non-owner update/remove with "Access denied".	M2	M1	SEC-QZ-01, SO-06
Functional requirements – Quizzes				
REQ-QZ-01	Create quiz with title, description, time limit, window and 1-50 questions.	M2	—	SEC-QZ-01
REQ-QZ-02	Reject invalid quiz, naming the failing field and rule.	M2	—	SEC-QZ-01, NFR-QUAL-03
REQ-QZ-03	Save quiz as Draft; visible to Students only after publishing.	M2	M4	SEC-QZ-01
REQ-QZ-04	Copy selected questions into the quiz at publication.	M2	M4	SEC-QZ-01
REQ-QZ-05	Edit Draft, delete with confirmation; reject once an attempt has started.	M2	M4	SEC-QZ-01
REQ-QZ-06	Student lists open and upcoming published quizzes without questions or answers.	M4	M2	—
REQ-QZ-07	Start attempt only inside the window and only once per quiz.	M4	M2	SEC-QZ-04, SO-06
REQ-QZ-08	Show questions and remaining time; server-side timer; no answers sent before submission.	M4	M2	SEC-QZ-02, SEC-QZ-04, SO-06, NFR-PERF-06
REQ-QZ-09	Select/change one option per question; warn on unanswered; record on submit.	M4	—	SEC-QZ-05
REQ-QZ-10	Auto-submit at time limit; reject late answers.	M4	—	SEC-QZ-04, SO-06
REQ-QZ-11	Auto-grade: 1 mark per correct answer, no negative marking; score and percentage.	M4	M2	SEC-QZ-05, SO-06, NFR-PERF-04
REQ-QZ-12	Grade and show result within 2 seconds (50 questions, 100 concurrent submissions).	M4	—	NFR-PERF-04
REQ-QZ-13	Show score, percentage and per-question answers, correct answers, explanation.	M4	M2	SEC-QZ-03
REQ-QZ-14	Submitted attempt cannot be changed or resubmitted.	M4	—	SEC-QZ-05, SO-06
REQ-QZ-15	Student views own attempts and detailed results.	M4	M3	SEC-QZ-03
REQ-QZ-16	Quiz owner views each Student's score, percentage and submission time.	M2	M4, M3	SEC-QZ-03
REQ-QZ-17	Role- and ownership-based access to quizzes, attempts and results.	M2	M4, M1	SEC-QZ-01, SEC-QZ-03, SO-06
REQ-QZ-18	Inform user when a quiz, question or attempt is unavailable; show no details.	M2	M4	NFR-QUAL-03
Functional requirements – Performance Analytics				
REQ-PA-01	Authenticated Student views completed attempts with quiz title, score, percentage and submission date.	M3	M4	SEC-QZ-03
REQ-PA-02	Calculate and display Student performance from completed quiz attempts.	M3	M4	—
REQ-PA-03	Allow an authenticated Student to view quiz performance grouped by topic.	M3	M2, M4	—
REQ-PA-04	Faculty views performance for Students who submitted attempts for quizzes owned by that Faculty.	M3	M2	SEC-QZ-03
REQ-PA-05	Display class performance including Student name, score, percentage and submission information.	M3	M2, M4	SEC-QZ-03
REQ-PA-06	Deny Faculty access to performance for a quiz not owned by that Faculty with "Access denied".	M3	M2, M1	SEC-QZ-03, SEC-ACC-01

ID	Existing requirement title / short description	Primary	Supporting	Related security / quality IDs
Functional requirements – Collaboration				
REQ-COL-01	Student creates a study group with required group information.	M3	—	SEC-ACC-01
REQ-COL-02	Student joins an existing study group.	M3	—	SEC-ACC-01
REQ-COL-03	Prevent duplicate study-group membership.	M3	—	—
REQ-COL-04	Student views the study groups to which they belong.	M3	—	SEC-ACC-01
REQ-COL-05	Admin moderates collaborative content.	M3	M1	SEC-ACC-01, BR-01
REQ-COL-06	Deny non-Admin moderation with "Access denied".	M3	M1	SEC-ACC-01
REQ-COL-07	Inform user when a study group or collaborative content is removed/unavailable without displaying unavailable details.	M3	—	—
Security objectives (SRS 3.3.3.1)				
SO-01	Account confidentiality	M1	—	—
SO-02	Material access control	M1	M4	—
SO-03	Integrity	M1	M4	—
SO-04	Secure authentication and sessions	M1	M2, M3, M4	—
SO-05	Secure communication	M1	M2, M3, M4	—
SO-06	Assessment integrity	M1	M2, M4	—
Security requirements – Account				
SEC-ACC-01	Require successful authentication before any protected account or material function.	M1	M2, M3, M4	—
SEC-ACC-02	Reject invalid credentials without disclosing which one was wrong.	M1	—	—
SEC-ACC-03	Store passwords only as salted bcrypt hashes; never return or log them.	M1	—	—
SEC-ACC-04	Terminate session on logout or 30-minute inactivity; reject later use of the token.	M1	—	—
SEC-ACC-05	Allow users to modify only their own account information.	M1	—	—
SEC-ACC-06	Lock account for 15 minutes after 5 consecutive failed logins.	M1	—	—
Security requirements – Material				
SEC-MAT-01	Deny material access to anyone who is neither owner nor authorized user.	M4	M1	—
SEC-MAT-02	Only the owner may share a material or revoke access to it.	M4	—	—
SEC-MAT-03	Only the owner may update or remove a material.	M4	—	—
SEC-MAT-04	Deny access on the next request after revocation or removal.	M4	—	—
Security requirements – Communication				
SEC-COM-01	Transmit all client-server data over TLS 1.2 or higher; reject unencrypted connections.	M1	M2, M3, M4	—
Security requirements – Quizzes				
SEC-QZ-01	Only Faculty create, update or remove questions and quizzes.	M2	M1	—
SEC-QZ-02	No correct options or explanations to a Student before submission.	M4	M2	—
SEC-QZ-04	Server enforces availability window and time limit.	M4	M2	—
SEC-QZ-05	Attempts immutable after submission; scores computed only by the server.	M4	M2	—
SEC-QZ-03	Only the attempt owner views a result; only the quiz owner views submissions.	M2	M4, M3	—
Non-functional – Performance (3.3.1)				

ID	Existing requirement title / short description	Primary	Supporting	Related security / quality IDs
NFR-PERF-01	95% of page loads within 3 seconds with 200 concurrent users.	M1	M2, M3, M4	—
NFR-PERF-02	Support 200 concurrent users with fewer than 1% failed requests.	M1	M2, M3, M4	—
NFR-PERF-03	Material share or revocation takes effect within 2 seconds.	M4	—	—
NFR-PERF-04	Grading complete and result available within 2 seconds.	M4	M2	—
NFR-PERF-05	Question-bank search within 2 seconds for up to 1,000 questions.	M2	—	—
NFR-PERF-06	Quiz questions displayed within 3 seconds of Start Quiz.	M4	M2	—
Non-functional – Safety (3.3.2)				
NFR-SAFE-01	Daily backups; restore from a backup no more than 24 hours old.	M1	M4	—
NFR-SAFE-02	Interrupted upload, update or removal leaves the previous material state unchanged.	M4	—	—
NFR-SAFE-03	Downloaded file is identical to the uploaded file (checksum).	M4	—	—
Non-functional – Quality attributes (3.3.4)				
NFR-QUAL-01	First-time user can register, log in and upload within 5 minutes without instructions.	M1	M4	—
NFR-QUAL-02	Availability of at least 99% per month, excluding announced maintenance.	M3	M1, M2, M4	—
NFR-QUAL-03	Invalid input or unsupported files rejected without crashing or affecting other users.	M1	M2, M3, M4	—
NFR-QUAL-04	Works on current Chrome, Firefox, Edge and Safari on desktop and mobile.	M1	M2, M3, M4	—
NFR-QUAL-05	Features implemented as separate, independently testable modules.	M3	M1, M2, M4	—
NFR-QUAL-06	Every requirement has a unique ID and is traceable to at least one test case.	M3	M1, M2, M4	—
Business rules (3.3.5)				
BR-01	Protected functions are available only to authenticated users.	M1	M2, M3, M4	—
BR-02	Each study material has exactly one owner: the user who uploaded it.	M4	—	—
BR-03	Only the owner may share, update, remove or revoke access to a material.	M4	—	—
BR-04	A user may view or download a material only if they are its owner or have been granted access.	M4	—	—
BR-05	Materials may be shared only with registered users.	M4	M1	—
BR-06	Users may change only their own profile; email address and role cannot be changed.	M1	—	—
BR-07	Users may upload only material they have the right to share. The uploader is responsible for copyright compliance.	M4	—	—
BR-08	Admin accounts are created by platform administrators, not through public registration.	M1	M3	—
Domain requirement (SRS 3.3.5, unnumbered)				
(no ID)	All user-facing content and messages are in English.	M3	M1, M2, M4	—

Titles are the brief descriptions already used in Appendix C (functional, security and non-functional requirements), the objective names in 3.3.3.1 and the rule text in 3.3.5. M1–M4 denote Members 1–4. The domain requirement has no ID in the original SRS; none has been invented.

D.4 Cross-Module Coordination Arrangements

Requirement(s)	Coordination arrangement
REQ-ACC-10, SEC-ACC-01, DC-07	M1 provides the authentication and role/ownership middleware; M2, M3 and M4 apply it to their own endpoints and each verify it in their own module tests.
REQ-QB-07, REQ-QZ-17, REQ-MAT-10, REQ-PA-06, REQ-COL-06	Access-control requirements of each module: the module owner is primary; M1 is supporting as owner of the middleware that enforces role and ownership checks.
REQ-QZ-03, 04, 05	M2 owns Draft/Publish, the question snapshot and edit/delete rules; M4 consumes the snapshot and supplies the "attempt started" state.
REQ-QZ-17, REQ-QZ-18, SEC-QZ-03	M2 is primary and coordinates; M4 implements and tests the Student/attempt-side clauses; M1 supplies role checks.
REQ-QZ-15 and REQ-PA-01; REQ-QZ-16 and REQ-PA-04, REQ-PA-05	Overlapping requirements with different owners. M4 (attempt records) and M3 (performance views) agree one shared attempt-query interface; M2 and M3 agree the quiz-owner view. Behaviour is unchanged.
SO-01 to SO-06, SEC-COM-01	M1 coordinates security verification across modules; module owners implement and test the linked requirements in their own modules.
NFR-PERF-01, NFR-PERF-02, NFR-SAFE-01, NFR-QUAL-01, 03, 04	M1 coordinates (as in STP Section 9); M4 supports for material data, File Storage backup and the upload journey; all members remain responsible for their own modules.
NFR-QUAL-02, 05, 06	M3 coordinates availability, modularity and traceability verification (as in STP Section 9); all members maintain traceability rows for their own IDs.
REQ-ACC-09	M1 owns profile validation; M4 owns File Storage used for profile pictures.
All other items with a Supporting member in D.3	The primary owner agrees the interface with the listed supporting member(s) before implementation; the supporting member reviews the related tests. Responsibility for the requirement stays with the primary owner.

D.5 Summary of Primary Ownership

Member	Functional (REQ)	Security req. (SEC)	Non-functional (NFR)	Security obj. (SO)	Business rules (BR)	Total
M1	10	7	6	6	3	32
M2	15	2	1	0	0	18
M3	13	0	3	0	0	16
M4	21	7	5	0	5	38
Total	59	16	15	6	8	104

Counts are a coverage check, not a workload measure: they cover 59 functional requirements, 16 security requirements, 15 non-functional requirements, 6 security objectives and 8 business rules (104 items). Workload balance between members is an open team decision.

D.6 Design Constraints – Coordination Owners

ID	Design constraint (SRS 3.4)	Coordinator	Supporting
DC-01	The system shall be a web application built on the MERN stack (MongoDB, Express, React, Node.js).	M1	M2, M3, M4
DC-02	The client shall communicate with the server only through a defined API, specified in the Software Architecture and Design Specification.	M2	M1, M3, M4
DC-03	All client-server communication shall use HTTPS with TLS 1.2 or higher (SEC-COM-01).	M3	M1, M2, M4
DC-04	Passwords shall be stored only as salted bcrypt hashes (SEC-ACC-03).	M1	—
DC-05	The client shall run on the browsers listed in NFR-QUAL-04.	M1	M2, M3, M4
DC-06	The file types and size limits (PDF, DOCX, PPTX, PNG; 25 MB), field length limits (title 100, description 500 characters) and profile picture limit (2 MB) defined in REQ-MAT-01, REQ-MAT-02 and REQ-ACC-09 shall be enforced by the storage and validation design.	M4	M1
DC-07	Session, role and ownership checks shall be performed on the server for every protected request, and shall not depend on client-side controls alone.	M1	M2, M3, M4
DC-08	Session expiry and account lockout (REQ-ACC-06, REQ-ACC-07) shall be enforced on the server.	M1	—
DC-09	The implementation shall stay within the scope of Account Management, Material Sharing, Question Bank & Quizzes, and Performance & Collaboration.	M3	M1, M2, M4

ID	Design constraint (SRS 3.4)	Coordinator	Supporting
DC-10	The requirement IDs in this SRS shall be kept unchanged in the Software Architecture and Design Specification and the Test Plan.	M2	M1, M3, M4
DC-11	The system shall be developed by the four-member project team within the course schedule.	M4	M1, M2, M3

Design constraints are listed for completeness; they are not counted in D.5. DC-11 is shown as amended in revision 1.1.